



ETHYLENE RESPONSE FACTORS 4.1/4.2 with an EAR motif repress anthocyanin biosynthesis in red-skinned pears

Hongye Sun ¹, Kangdi Hu ¹, Shuwei Wei,² Gaifang Yao ^{1,*} and Hua Zhang ^{1,*}

¹ School of Food and Biological Engineering, Hefei University of Technology, Hefei 230009, China

² Shandong Institute of Pomology, Tai'an 271000, China

*Author for correspondence: 2017800495@hfut.edu.cn (G.Y.), hzhanglab@hfut.edu.cn (H.Z.).

The author responsible for distribution of materials integral to the findings presented in this article in accordance with the policy described in the Instructions for Authors (<https://academic.oup.com/plphys>) is Hua Zhang (hzhanglab@hfut.edu.cn).

Abstract

Red-skinned pears (*Pyrus* L.) are preferred to consumers for their attractive color and abundant anthocyanins. *Pyrus* ETHYLENE RESPONSE FACTOR 3 (PyERF3) positively regulates anthocyanin biosynthesis through interacting with *Pyrus* myeloblastosis family 114 (PyMYB114) and *Pyrus* basic helix-loop-helix 3 (PybHLH3) in red-skinned pears. However, the role of APETALA2/ethylene response factors (AP2/ERFs), which negatively regulate anthocyanin biosynthesis, remains unclear in red-skinned pears. Here, we validated that 2 AP2/ERFs, PyERF4.1 and PyERF4.2, screened from the transcriptome data of 'Starkrimson' pear (*Pyrus communis* L.) and its green mutant, inhibit anthocyanin biosynthesis in transgenic pear calli, as well as in overexpression and gene-edited tomato (*Solanum lycopersicum*) fruits. Meanwhile, the co-transformation of PyERF4.1/PyERF4.2 with PyERF3–PyMYB114–PybHLH3 inhibited anthocyanin biosynthesis in pear fruits and strawberry (*Fragaria vesca*) receptacles. Further assays showed that PyMYB114 activated the transcription of PyERF4.1/PyERF4.2; PyERF4.1/PyERF4.2 then interacted with PyERF3 to affect the stability of the PyERF3–PyMYB114–PybHLH3 complex, thereby inhibiting the transcription of the anthocyanin biosynthesis gene *Pyrus anthocyanidin synthase* (PyANS). Furthermore, deletion of the ERF-associated-amphiphilic repression (EAR) motif eliminated the inhibitory effect of PyERF4.1/PyERF4.2 on anthocyanin biosynthesis, and a mutation of the PyERF4.2-EAR motif (LxLxM to LxLxL) strengthened the inhibitory effect, demonstrating that the EAR motif is indispensable for the inhibitory effect of PyERF4.1/PyERF4.2 on anthocyanin biosynthesis in pears. Our study has shed light on a feedback regulatory loop mechanism that balances the excessive accumulation of anthocyanins in red-skinned pears, providing insights into the regulatory mechanism of anthocyanin biosynthesis and the regulatory network of coloration in red-skinned pears.

Introduction

Pear (*Pyrus* L.) is among the most popular worldwide fruits, and red-skinned pears are preferred to consumers for their attractive color and abundant anthocyanins (Li et al. 2020). Anthocyanins are widely found in plants, and they not only enhance the dissemination of plants, but also have a major role in stress responses, such as preventing plants from ultraviolet damage, scavenging free radicals, and increasing

antioxidant activity (Peng et al. 2017). Furthermore, anthocyanins may also play an important role on humans against neurological and cardiovascular diseases (Yang et al. 2017).

Anthocyanins are biosynthesized by the flavonoid pathway (Koes et al. 2005; Hichri et al. 2011), where phenylalanine ammonia-lyase (PAL), chalcone synthase (CHS), chalcone isomerase (CHI), flavanone-3-hydroxylase (F3H), dihydroflavonol 4-reductase (DFR), anthocyanidin synthase (ANS), and

UDP-glucose: flavonoid 3-glucosyl transferase (UFGT) are key enzymes (Tanaka et al. 2008; Zhao and Dixon 2009). Many transcription factors (TFs) are involved in regulating plant anthocyanin biosynthesis. Myeloblastosis family (MYB) TFs, basic helix-loop-helix (bHLH) TFs, and tryptophan-aspartic acid repeat (WDR) TFs regulated anthocyanin biosynthesis in different plants (Hichri et al. 2010; Xu et al. 2014). In general, anthocyanins are controlled mainly by the MYB–bHLH–WD40 (MBW) complex which composed of at least 3 TFs, belonging to 3 different families (Ramsay and Glover 2005; Liu et al. 2021). MYBs, as central regulators in the MBW complex, have been widely reported to regulate anthocyanin biosynthesis individually or together with bHLHs in horticultural plants. For example, overexpression of *FvMYB10* in strawberry (*Fragaria vesca*) and *SlMYB75* in tomato (*Solanum lycopersicum*) fruits substantially promoted anthocyanin biosynthesis (Jian et al. 2019; Castillejo et al. 2020). In apple (*Malus domestica*), *MdMYB10* enhances anthocyanin synthesis by interacting with *MdbHLH3* and *MdbHLH33* (Espley et al. 2007). In pear, *PyMYB114*, *PyMYB10* or *PyMYB10b* interacted with *PybHLH3* and substantially enhanced anthocyanin biosynthesis (Zhai et al. 2016; Yao et al. 2017). Moreover, peach (*Prunus persica*) *PpMYB18* and red-skinned pear *PyMYB140* act as repressors competing with anthocyanin-associated MYB activators to bind bHLHs for regulating anthocyanin accumulation (Zhou et al. 2019; Ni et al. 2021).

There is increasing evidence that in addition to the MBW complex, the APETALA2/ethylene response factor (AP2/ERF) family also participates in regulating anthocyanin biosynthesis (An et al. 2020a, 2020b; Ma et al. 2021). AP2/ERFs are a large family of plant-specific TFs with a conserved AP2 domain and have been found to be involved in plant developmental processes, like regulating plant secondary metabolite synthesis, and various stress responses via binding to the short cis-acting elements such as the GCC-box and dehydration responsive elements/C-repeat element motifs in the target gene promoter (Liu et al. 1998; Fujimoto et al. 2000; Li et al. 2018). Generally, ERF proteins have no specific sequence motifs as transcriptional activators but tend to be enriched in acidic amino acids; however, as repressors, their C-terminal region usually contain an ERF-associated-amphiphilic repression (EAR) motif defined by the consistent sequence patterns of either LxLxL or DLNLP (Ohta et al. 2001; Li et al. 2018; Yang et al. 2018). EAR motif-mediated transcriptional repression is the major form of transcriptional repression found in plants to date (Kagale et al. 2010; Choi et al. 2018).

A number of studies have revealed that AP2/ERFs participate in regulating anthocyanin biosynthesis by interacting with the MBW complex, and they have diverse regulatory models of anthocyanin biosynthesis in different plant species under different conditions (Yao et al. 2017; An et al. 2018; Wu et al. 2020; Ni et al. 2021). In phytohormone-induced anthocyanin biosynthesis, *MdERF1B* and *MdERF3* directly activate the expression of *MdMYB11* and *MdMYB1*, respectively, thereby promoting ethylene-induced anthocyanin biosynthesis in apple fruits

(An et al. 2018; Zhang et al. 2018); jasmonate and ethylene-regulated *PyERF22* enhances the activation of the *PyUFGT* promoter by interacting with *PyMYB10* and *PyMYB10b* to promote lanolin-induced anthocyanin biosynthesis in 'Zaosu' pear fruits (*Pyrus pyrifolia* Nakai) (Wu et al. 2020); ethylene-activated *PyERF105* induces the repressor *PyMYB140* expression for inhibiting anthocyanin biosynthesis in red-skinned pear fruits (Ni et al. 2021). In light-induced anthocyanin biosynthesis, *Py4ERF24* and *Py12ERF96* are verified to enhance blue light-induced anthocyanin biosynthesis by interacting with *PyMYB114* in 'Hongzaosu' pear fruits (*P. pyrifolia* Nakai) (Ni et al. 2019). In drought stress-induced anthocyanin biosynthesis, *MdERF38* interacts with *MdMYB1* to promote drought stress-induced anthocyanin biosynthesis in apple fruits (An et al. 2020b). Besides, *PyERF3* screened from 'Starkrimson' pear (*P. communis* L.) and its green mutant interacts with *PyMYB114* and *PybHLH3* to regulate anthocyanin biosynthesis (Yang et al. 2015; Yao et al. 2017). Thus, the above studies showed that many AP2/ERFs positively regulate anthocyanin biosynthesis, whereas a negative regulation of anthocyanin biosynthesis by repressor-type AP2/ERFs in horticultural plants is unclear. Meanwhile, the color of red-skinned pears is often unstable and sometimes does not develop, which seriously restricts the development of red-skinned pear industry (Ni et al. 2021). Hence, it is of great importance and necessity to investigate how AP2/ERFs inhibit anthocyanin biosynthesis and thus balance the excessive accumulation of anthocyanins in red-skinned pears.

In this study, 2 AP2/ERFs, *PyERF4.1* and *PyERF4.2*, containing EAR motifs were screened from the transcriptome data of 'Starkrimson' pear and its green mutant during 3 fruit developmental stages by bioinformatics and correlation analysis. The inhibitory function of *PyERF4.1* and *PyERF4.2* on anthocyanin biosynthesis was verified in transgenic pear calli, stable overexpressed tomato fruits, and deletion mutation tomato fruits obtained by clustered regularly interspaced short palindromic repeats/CRISPR associated nuclease 9 (CRISPR/Cas9) editing. Co-expression of *PyERF4.1/PyERF4.2* with *PyERF3–PyMYB114–PybHLH3* in strawberry receptacles and pears by the transient expression system substantially reduced anthocyanins accumulation. Further analysis showed that *PyMYB114* binds to the *PyERF4.1* or *PyERF4.2* promoters and activates their transcription, then *PyERF4.1* and *PyERF4.2* were verified to interact with *PyERF3* through EAR motif and this interaction results in decreased stability of the *PybHLH3–PyMYB114–PyERF3* complex, thereby repressing the anthocyanin biosynthesis gene *Pyrus anthocyanidin synthase* (*PyANS*) transcription. Besides, the integrity of the EAR motif of *PyERF4.1* and *PyERF4.2* determines the strength of the inhibitory effect. Therefore, our data provide insights into the inhibitory mechanism of *PyERF4.1* and *PyERF4.2* for anthocyanin biosynthesis in red-skinned pears, which could help to refine the network of regulation for anthocyanin biosynthesis in pears.

Results

Screening of AP2/ERF candidate genes by transcriptome data and bioinformatics analysis

To explore the regulatory mechanism of anthocyanin biosynthesis in red-skinned pears by AP2/ERFs, deep exploration of transcriptome data of red-skinned pear ‘Starkrimson’ and its green mutant revealed 114 AP2/ERFs were differentially expressed genes at 40, 55, and 85 days after full bloom (DAFB) (Yang et al. 2015). Among of them, 33 AP2/ERFs were downregulated in ‘Starkrimson’ pear comparing with its green mutant by heat map analysis (Fig. 1A). Next, 33 AP2/ERFs were compared with the reported negative regulators associated with color changes in other species (SIAP2a,

SIERF6, MdERF1, MdERF2, AdERF9, MaERF11, MdERF4, and EjERF11) and a phylogenetic tree was constructed using MEGA7 software and the neighbor-joining method with bootstrap analysis (1,000 replicates). As shown in Fig. 1B, Pbr000398.1, Pbr038280.1, and Pbr016222.1 were most closely related to the evolution of color change-related negative regulators EjERF11, MdERF4, MaERF11, and AdERF9, and expression of their genes were substantially downregulated during the critical period of red-skinned pear peel coloration, so Pbr000398.1, Pbr038280.1, and Pbr016222.1 were selected for the further study.

Furthermore, amino acid sequence alignment showed the presence of the AP2 domain in all these proteins, and it was found that Pbr000398.1, EjERF11, MdERF4, MaERF11, and

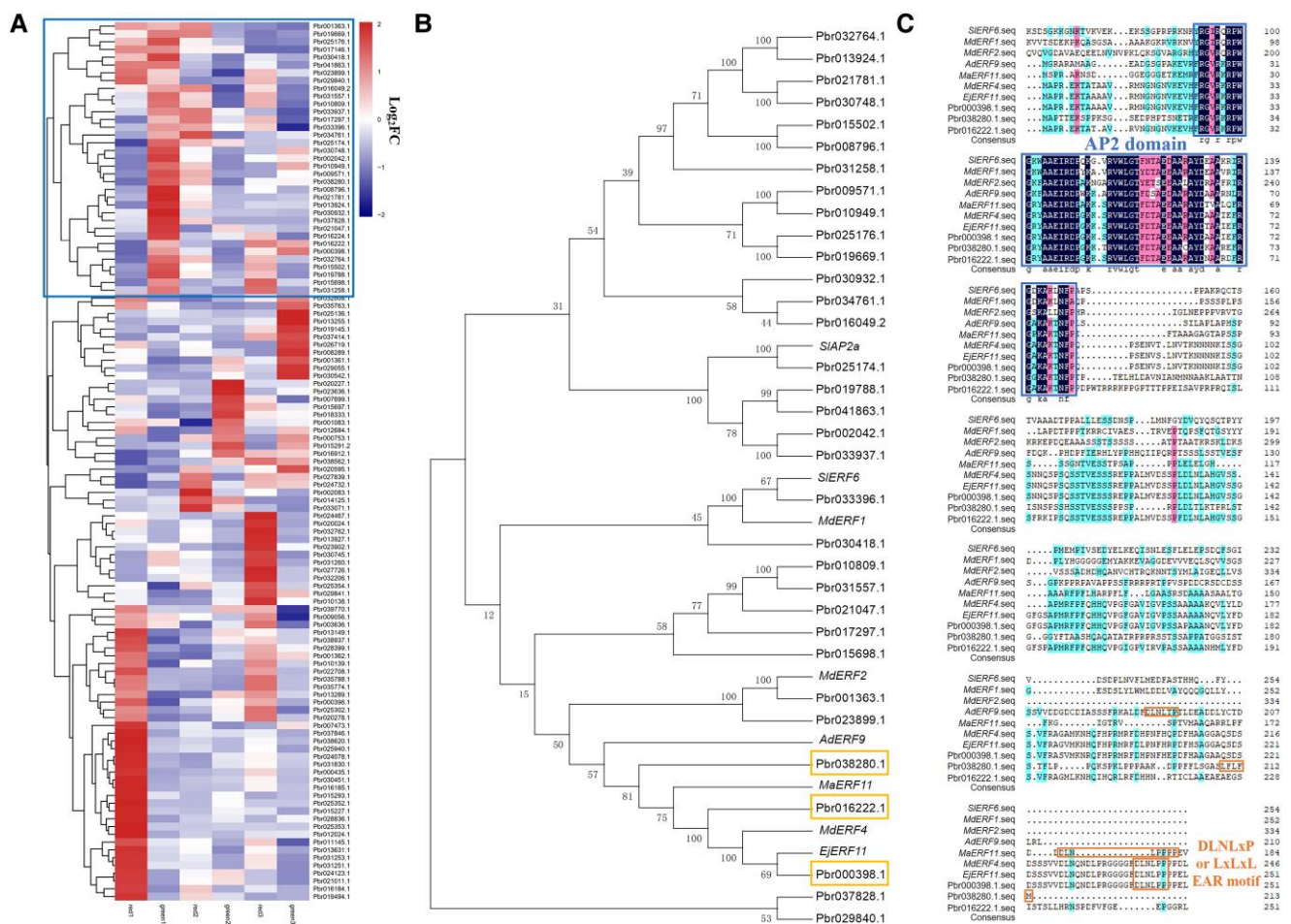


Figure 1. The candidate AP2/ERF genes screened from transcriptome data of ‘Starkrimson’ pear and its green mutant bioinformatics analysis. A) Heat map of 114 AP2/ERF genes differentially expressed at different developmental stages in the peel of ‘Starkrimson’ pear and its green mutant. Red1, red2, and red3 are the samples of ‘Starkrimson’ red-skinned pear at 40, 55 and 85 DAFB; green1, green2, and green3 denote the samples of the green mutant of ‘Starkrimson’ at 40, 55 and 85 DAFB, respectively. The value (Log₂FC) of each ERF gene was calculated by comparing the expression of each gene in ‘Starkrimson’ pear to its green mutant. The square box shows the AP2/ERFs screened for downregulated expression in red pears. B) The downregulated expression of 41 AP2/ERFs in different plant species are analyzed in the phylogenetic tree. The C-terminus of the ERF domain was analyzed by the MEGA7 Program using the neighbor-joining method. The figures next to the branches indicated the percentage of replicated trees in which the relevant taxa clustered together in the bootstrap test (1,000 replications). The yellow boxes show Pbr000398.1, Pbr038280.1, Pbr016222.1, respectively. C) The amino acid sequence alignment analysis of the candidate AP2/ERFs. EAR motif, ERF-associated-amphiphilic repression motif. The blue box shows the protein sequence of AP2 domain, and the orange box shows the protein sequence and location of the EAR negative repressor motif.

AdERF9 contained an intact complete EAR motif (DLNLxP), whereas Pbr038280.1 had an incomplete EAR motif with a mutated amino acid (LxLxM), Pbr016222.1 had no EAR motif (Fig. 1C). Thus, it could be hypothesized that Pbr000398.1 and Pbr038280.1 negatively regulate anthocyanin biosynthesis in red-skinned pears, and the expression levels of their genes were substantially downregulated in ‘Starkrimson’ pears compared with the green mutant at different developmental stages. Therefore, Pbr000398.1 and Pbr038280.1 were screened for further investigation and named PyERF4.1 and PyERF4.2, respectively.

Negative correlation of PyERF4.1/PyERF4.2 expression with anthocyanin biosynthesis in red-skinned pears

For exploring and verifying associations of PyERF4.1/PyERF4.2 with the structural genes and TFs involved in anthocyanin biosynthesis, the anthocyanin content of red-skinned pears ‘Hongzaosu’, ‘Starkrimson’, and green-skinned pears ‘Zaosu’, ‘Jinzheng NO.1’ were measured at 30, 60 and 90 DAFB. The results showed that total anthocyanin contents of ‘Hongzaosu’ and ‘Starkrimson’ pear peels were the highest at 30 DAFB, then decreased with fruit development, and they were always higher than those of ‘Zaosu’ and ‘Jinzheng NO.1’ pear peels ($P < 0.01$) (Fig. 2A; Supplemental Fig. S1A). In addition, PyERF4.1 and PyERF4.2 expression levels were much higher in ‘Zaosu’ and ‘Jinzheng NO.1’ pear peels than in ‘Hongzaosu’ and ‘Starkrimson’ pear peels and increased with fruit development, while PyERF3, PyMYB114, PybHLH3, PyDFR, PyANS, and PyUGT expression levels showed opposite patterns (Fig. 2, B–I; Supplemental Fig. S1, B–I). The correlation analysis showed that PyERF4.1 and PyERF4.2 expression levels were substantially negatively correlated with anthocyanins content, the expression levels of PyERF3, PyMYB114, PybHLH3, and the expression levels of anthocyanin biosynthesis genes PyDFR, PyANS and PyUGT (Fig. 2J). This suggested that PyERF4.1 and PyERF4.2 may negatively regulate anthocyanin biosynthesis in red-skinned pears.

PyERF4.1 and PyERF4.2 negatively regulate anthocyanin biosynthesis in pears

To verify the function of PyERF4.1 and PyERF4.2 in regulating anthocyanin biosynthesis, PyERF4.1 and PyERF4.2 were overexpressed in pear calli. The obtained transgenic pear calli and wild-type (WT) pear calli were treated with light, and then total anthocyanin contents were measured. After 15 d of light treatment, WT pear calli produced abundant anthocyanins, PyERF4.1-OE transgenic calli had no anthocyanin accumulation, while PyERF4.2-OE transgenic calli produced a little anthocyanin (Fig. 3A). The reverse transcription quantitative PCR (RT-qPCR) analysis showed that PyERF4.1 and PyERF4.2 were overexpressed in transgenic pear calli, respectively (Fig. 3, C and D). Compared to the WT calli, PyERF4.1-OE and PyERF4.2-OE calli showed significantly lower anthocyanin contents and a^* values ($P < 0.05$) (Fig. 3, B and E). In

PyERF4.1-OE and PyERF4.2-OE calli, PyERF3, PyMYB114, PyDFR, PyANS, and PyUGT expression levels were significantly reduced compared to WT calli ($P < 0.05$), and their expression levels were more reduced in PyERF4.1-OE calli than in PyERF4.2-OE calli (Fig. 3F). In summary, PyERF4.1 and PyERF4.2 negatively regulated anthocyanin biosynthesis in pear calli, and the inhibitory effect of PyERF4.1 was stronger than that of PyERF4.2.

In order to further verify the inhibitory function of PyERF4.1/PyERF4.2 in anthocyanin biosynthesis of red-skinned pears, PyERF4.1 and PyERF4.2 were transiently silenced in ‘Zaosu’ pears. For avoiding silencing other genes with similar sequences, specific coding fragments from the 3’ region of the TFs were selected and inserted into the pSAK277 vector, resulting in 2 constructs: PyERF4.1-RNAi and PyERF4.2-RNAi. In Fig. 3G, there was no pigmentation observed in the empty vector pSAK277, PyERF4.1 and PyERF4.2 transformed pear peels, while obvious pigmentations were observed in PyERF4.1-RNAi and PyERF4.2-RNAi transformed pear peels. Compared with the control pear peels, PyERF4.1 and PyERF4.2 expression levels were significantly reduced ($P < 0.05$) (Fig. 3, I and J) and PyERF3, PyMYB114, PyDFR, PyANS, and PyUGT expression levels were significantly increased ($P < 0.05$) in PyERF4.1-RNAi and PyERF4.2-RNAi transformed pear peels as shown by RT-qPCR analysis (Fig. 3, L–P). In addition, total anthocyanin contents and a^* value were also significantly elevated when PyERF4.1-RNAi and PyERF4.2-RNAi were transformed ($P < 0.05$) (Fig. 3, H and K), which was consistent with the phenotype. Therefore, these results further validated the function of PyERF4.1 and PyERF4.2 in inhibiting anthocyanin biosynthesis in pears.

PyERF4.1 negatively regulates anthocyanin biosynthesis in tomato fruits

Due to the stronger inhibitory effect of PyERF4.1 than PyERF4.2, PyERF4.1 was overexpressed in tomato fruits and T2 generation PyERF4.1 overexpressing (ERF4.1-OE) transgenic tomato plants were obtained (Supplemental Fig. S2). However, for the difficulties associated with stable pear transformation, we screened the homologous gene *SIERF4.1* (C-terminus with a complete EAR motif) in tomato and obtained a T2 generation with the homologous mutation of tomato plants (*erf4.1*) by CRISPR/Cas9 (Supplemental Figs. S2 and S3). The above transgenic tomato lines were used to investigate the regulatory role of ERF4.1 on anthocyanin biosynthesis in tomato fruits.

RT-qPCR analysis revealed that PyERF4.1 was significantly overexpressed in ERF4.1-OE tomato fruits ($P < 0.05$) (Fig. 4A), while *SIERF4.1* expression was significantly reduced in *erf4.1* tomato fruits ($P < 0.05$) (Fig. 4B), suggesting the success of genetic manipulation. Compared to that of WT fruits, the anthocyanins content of ERF4.1-OE fruits was significantly lower by 1.5 times ($P < 0.05$), whereas *erf4.1* fruits had a significantly higher anthocyanins content by 1.4 times

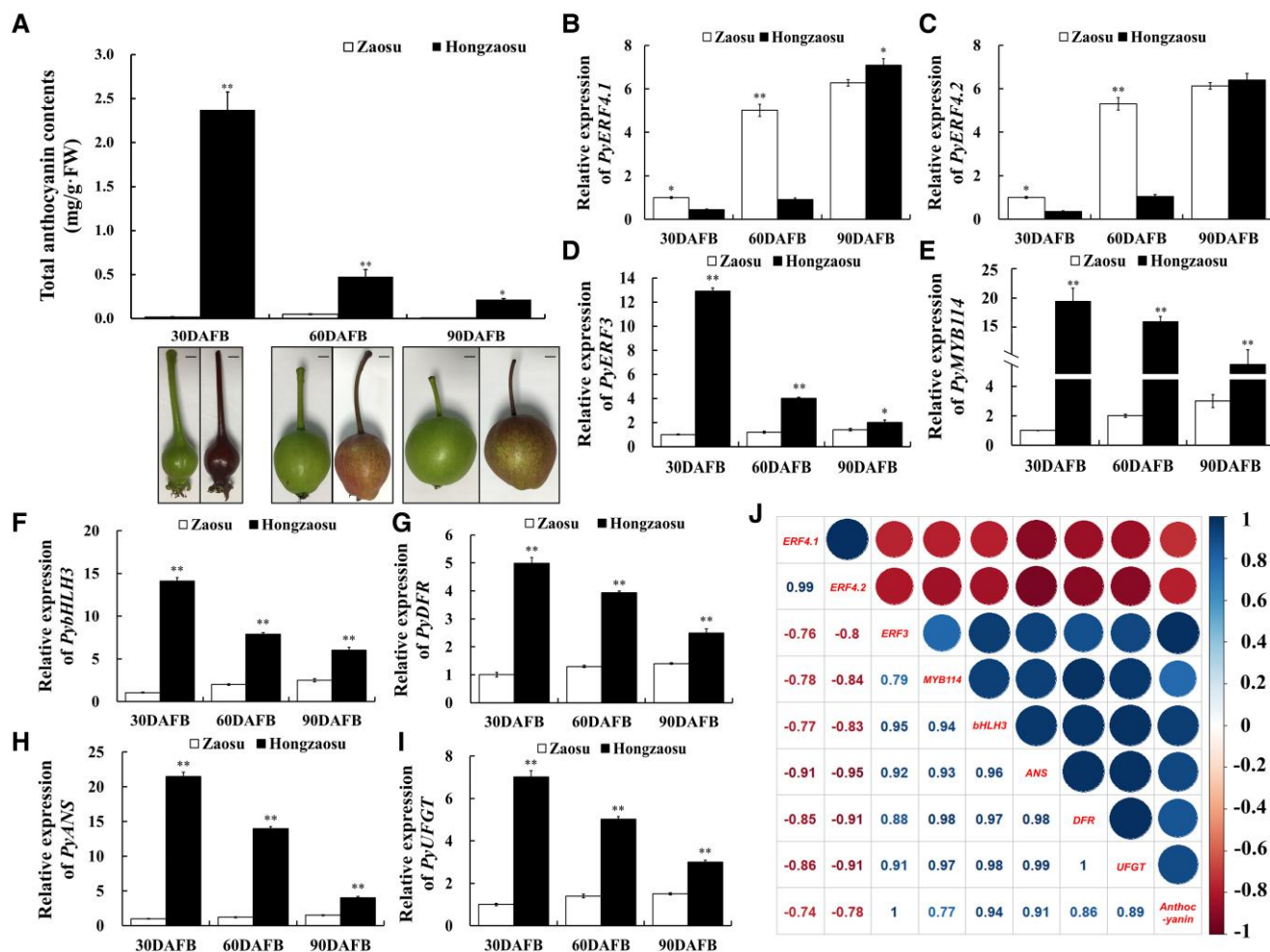


Figure 2. Assessing the correlation of candidate AP2/ERFs with anthocyanin biosynthesis-related genes in 'Zaosu' and 'Hongzaosu' pear cultivars. A) The appearance and total anthocyanin contents of 'Zaosu' and 'Hongzaosu' pears at 30, 60 and 90 DAFB. DAFB, days after full bloom. Scale bars = 1 cm. B–F) Relative expression of *PyERF4.1*, *PyERF4.2*, *PyERF3*, *PyMYB114*, and *PybHLH3* in 'Zaosu' and 'Hongzaosu' pears at 30, 60 and 90 DAFB. G–I) Relative expression of *PyDFR*, *PyANS*, and *PyUGT* in 'Zaosu' and 'Hongzaosu' pears at 30, 60 and 90 DAFB. J) Correlation analysis among the anthocyanin contents and gene expression levels of *PyERF4.1*, *PyERF4.2*, *PyERF3*, *PyMYB114*, *PybHLH3*, *PyDFR*, *PyANS*, and *PyUGT*. R scripts were used to analyze Pearson's correlation coefficients. "+" represents a positive correlation, "-" represents a negative correlation. Larger size and darker color of the circles mean stronger correlation. Bars indicate mean values $\pm$ SD from 3 biological replicates. Statistical analysis was carried out with Student's *t*-test, and significance was marked with asterisks (* $P < 0.05$) or (** $P < 0.01$).

($P < 0.05$) (Fig. 4C). *SIERF3*, *SIMYB114* expression levels and anthocyanin biosynthesis-related genes *SIDFR*, *SIANS*, *SIUGT* expression levels were significantly lower in *ERF4.1*-OE fruits than in WT fruits ($P < 0.01$), whereas they were significantly higher in *erf4.1* fruits than in WT fruits ($P < 0.01$) (Fig. 4D). In conclusion, *ERF4.1* negatively regulated anthocyanin biosynthesis in tomato fruits.

Co-transformation of *PyERF4.1*/*PyERF4.2* with the *PyERF3*–*PyMYB114*–*PybHLH3* complex substantially inhibits anthocyanin biosynthesis by transient expression system in pears

To investigate the inhibitory mechanism of *PyERF4.1* and *PyERF4.2* on anthocyanin biosynthesis in pears, a transient

transformation of *PyERF4.1*/*PyERF4.2* and related TFs (*PyERF3*, *PyMYB114*, and *PybHLH3*) which are known as activators of the anthocyanin pathway, was performed in pear peels. Different ratios of *PyERF4.1*/*PyERF4.2*:*PyERF3* (1:1, 2:1, 1:2) and *PyMYB114*–*PybHLH3* were co-transformed into 'Zaosu' pear peels, and the phenotype was observed 6 d after injection. In Fig. 5A, there was no pigmentation seen in the empty vector pSAK277-transformed peels. Some pigmentation was observed when *PyMYB114*–*PybHLH3* were cotransformed, while substantial pigmentation was observed when *PyERF3*–*PyMYB114*–*PybHLH3* were cotransformed. Moreover, when the transformation ratio of *PyERF4.1*/*PyERF4.2* to *PyERF3* increased from 1:1 to 2:1, the pigment deposition gradually decreased, while when the transformation percentage of *PyERF3* increased, the pigment accumulation

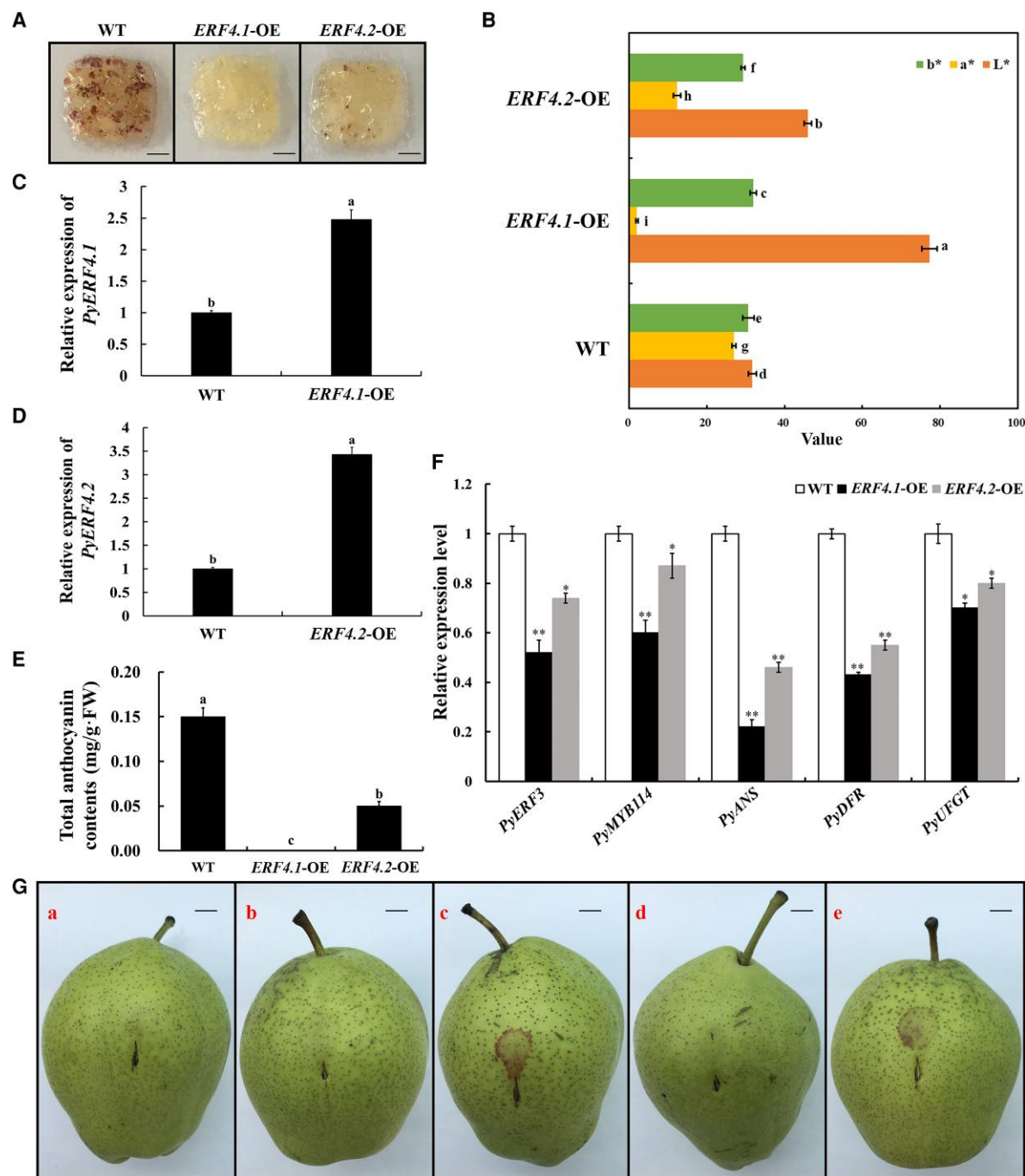


Figure 3. Validation of the inhibitory effect of PyERF4.1 and PyERF4.2 on anthocyanin biosynthesis in pears. A) The phenotype of PyERF4.1 and PyERF4.2 overexpressed pear calli after 15 d of the light treatment. WT, wild type; OE, overexpression. Scale bars = 2 mm. B) The difference in color is indicated by the values of L*, a*, and b*. L* indicates luminance; a* indicates a range from green to magenta; b* indicates a range from yellow to blue. Bars indicate mean values $\pm$ SD from 6 biological replicates. C, D) Relative expression of PyERF4.1 and PyERF4.2 in transgenic pear calli. Data are presented as means $\pm$ SD ($n = 3$). E) Total anthocyanin contents of PyERF4.1 and PyERF4.2 overexpressed in pear calli. Bars indicate mean values $\pm$ SD from 3 biological replicates. F) Relative expression of PyERF3, PyMYB114, PyDFR, PyANS, and PyUGT in transgenic pear calli. Data are presented as means $\pm$ SD ($n = 3$). G) The phenotype of ‘Zaosu’ pear peels after infiltration with RNAi-induced gene silencing method: a, pSAK277; b, PyERF4.1; c, PyERF4.1-RNAi; d, PyERF4.2; and e, PyERF4.2-RNAi. The concentration and volume of Agrobacterium solution used were the same for each infiltration site. Scale bars = 1 cm. H) The difference in color is indicated by the values of L*, a*, and b*. L* indicates luminance; a* indicates a range from green to magenta; b* indicates a range from yellow to blue. Bars indicate mean values $\pm$ SD from 6 biological replicates. I, J) Relative expression of PyERF4.1 and PyERF4.2. Data are presented as means $\pm$ SD ($n = 3$). K) Total anthocyanin contents in transformed pear peels. Bars indicate mean values $\pm$ SD from 3 biological replicates. L–P) Relative expression of PyERF3, PyMYB114, PyDFR, PyANS, and PyUGT. Data are presented as means $\pm$ SD ($n = 3$). Statistical analysis was carried out with one-way ANOVA and Student’s *t*-test, and significance was marked with asterisks (* $P < 0.05$) or (** $P < 0.01$) or different letters ($P < 0.05$).

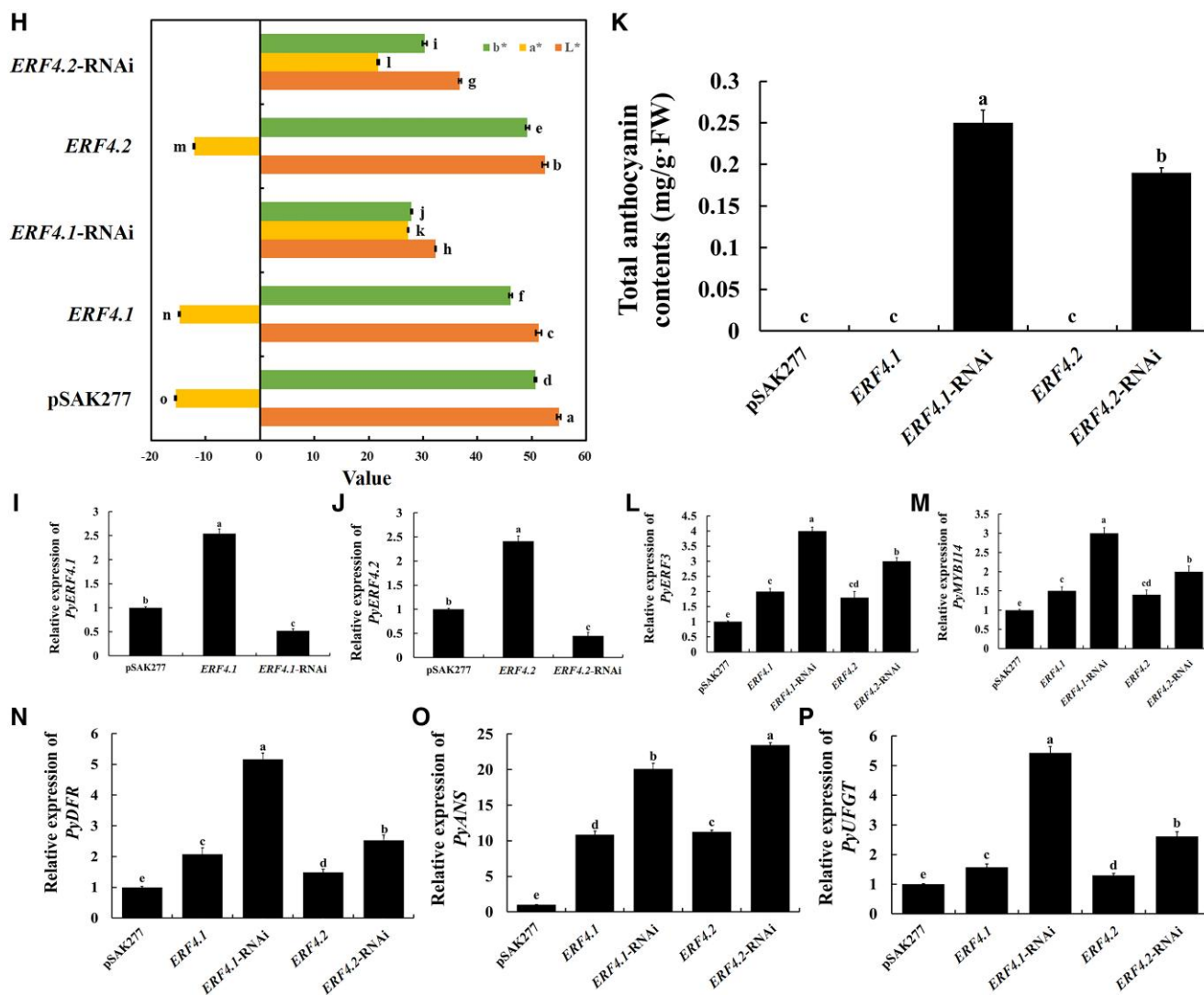


Figure 3. Continued

increased again. *PyERF4.1* still had a stronger inhibitory effect on pigment accumulation than *PyERF4.2*. With the increase of *PyERF4.1/PyERF4.2* transformation, total anthocyanin contents and a^* value decreased significantly ($P < 0.05$), which was consistent with the phenotypic changes after pear peels injection (Fig. 5, B and C).

Then, the major anthocyanin biosynthesis genes expression levels were analyzed in transfected pear peels by RT-qPCR. Consistent with the phenotypic results, *PyERF4.1/PyERF4.2* significantly reduced *PyDFR*, *PyANS*, and *PyUGT* expression levels after co-transformation with *PyERF3–PyMYB114–PybHLH3* ($P < 0.05$). Furthermore, this inhibitory effect enhanced as the amount of *PyERF4.1/PyERF4.2* transformation increased, while increasing the amount of *PyERF3* transformation could alleviate this inhibitory effect to some extent (Fig. 5D). Thus, *PyERF4.1/PyERF4.2* inhibits anthocyanin biosynthesis via potential interaction with *PyERF3–PyMYB114–PybHLH3* complex in pears.

Heterologous expression of *PyERF4.1/PyERF4.2* with a *PyERF3–PyMYB114–PybHLH3* complex inhibits anthocyanin accumulation in strawberry receptacles

For investigation of how *PyERF4.1/PyERF4.2* regulates anthocyanin biosynthesis with *PyERF3*, *PyMYB114*, and *PybHLH3*, a transient transformation of their genes into strawberry receptacles was performed. The pigmentation appeared at the infiltration sites 6 d after injection, and the change trend was similar to that of pear peels. In Fig. 6A, there was no pigmentation observed in the empty vector pSAK277-transformed receptacles. Slight pigmentation was observed when *PyMYB114–PybHLH3* were cotransformed, while substantial pigmentation was observed when *PyERF3–PyMYB114–PybHLH3* were cotransformed. In contrast, when *PyERF4.1/PyERF4.2* were cotransformed with *PyMYB114–PybHLH3* or *PyERF3–PyMYB114–PybHLH3*, substantial inhibition of pigmentation was observed, and *PyERF4.1* had a more pronounced inhibitory effect than *PyERF4.2*. In addition,

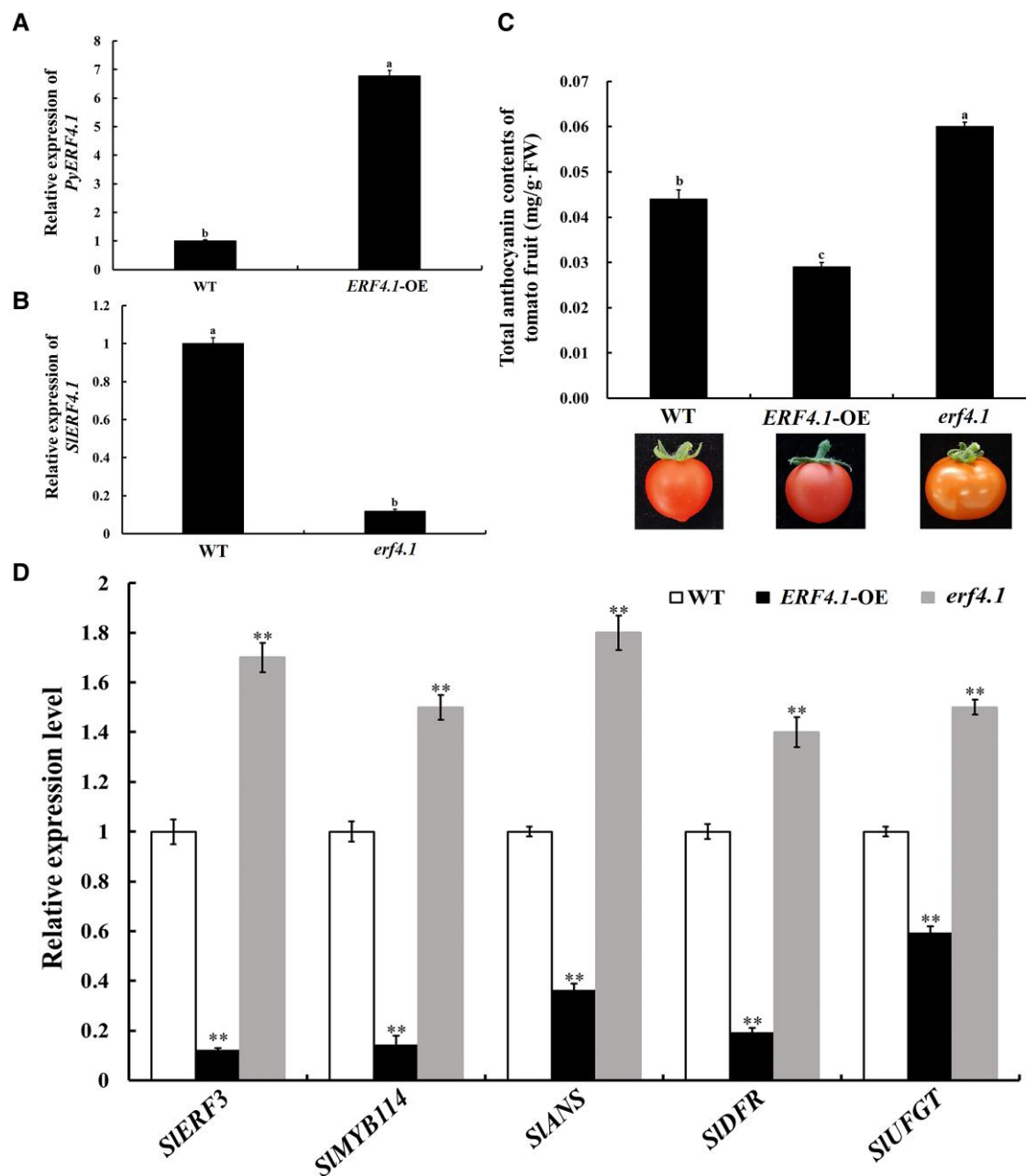


Figure 4. Validation of the inhibitory effect of PyERF4.1 on anthocyanin biosynthesis in transgenic tomato fruits. A, B) Relative expression of *PyERF4.1* and *SlERF4.1* in transgenic tomato fruits. WT, wild type; OE, overexpression; *erf4.1*, *ERF4.1* deletion mutant. Data are presented as means $\pm$ SD ($n = 3$). C) Total anthocyanin contents of *ERF4.1-OE* and *erf4.1* deletion mutant tomato fruits at red-ripening stage. Bars indicate mean values $\pm$ SD from 3 biological replicates. D) Relative expression of *SlERF3*, *SlMYB114*, *SlIDR*, *SlANS*, and *SlUGT* in transgenic tomato fruits. Data are presented as means $\pm$ SD ($n = 3$). Statistical analysis was carried out with one-way ANOVA and Student's *t*-test, and significance was marked with asterisks (* $P < 0.05$) or (** $P < 0.01$) or different letters ($P < 0.05$).

differences in strawberry receptacles color affect L^* , a^* , and b^* values (Fig. 6B). When *PyERF3*–*PyMYB114*–*PybHLH3* were co-transformed, anthocyanin contents in strawberry receptacles were significantly higher than when *PyMYB114*–*PybHLH3* were co-transformed ($P < 0.05$), while when *PyERF4.1*/PyERF4.2 were co-transformed with *PyMYB114*–*PybHLH3* or *PyERF3*–*PyMYB114*–*PybHLH3*, the anthocyanin contents were significantly lower ($P < 0.05$) (Fig. 6C). Meanwhile, the

anthocyanin content of strawberry receptacles cotransformed with *PyERF4.1* was always lower than that of strawberry receptacles cotransformed with *PyERF4.2* ($P < 0.05$). In conclusion, co-transformation of *PyERF4.1*/PyERF4.2 with *PyERF3*–*PyMYB114*–*PybHLH3* substantially reduced anthocyanin biosynthesis in strawberry receptacles.

Next, RT-qPCR analysis showed that *PyERF4.1*/PyERF4.2 significantly reduced anthocyanin biosynthesis-related genes

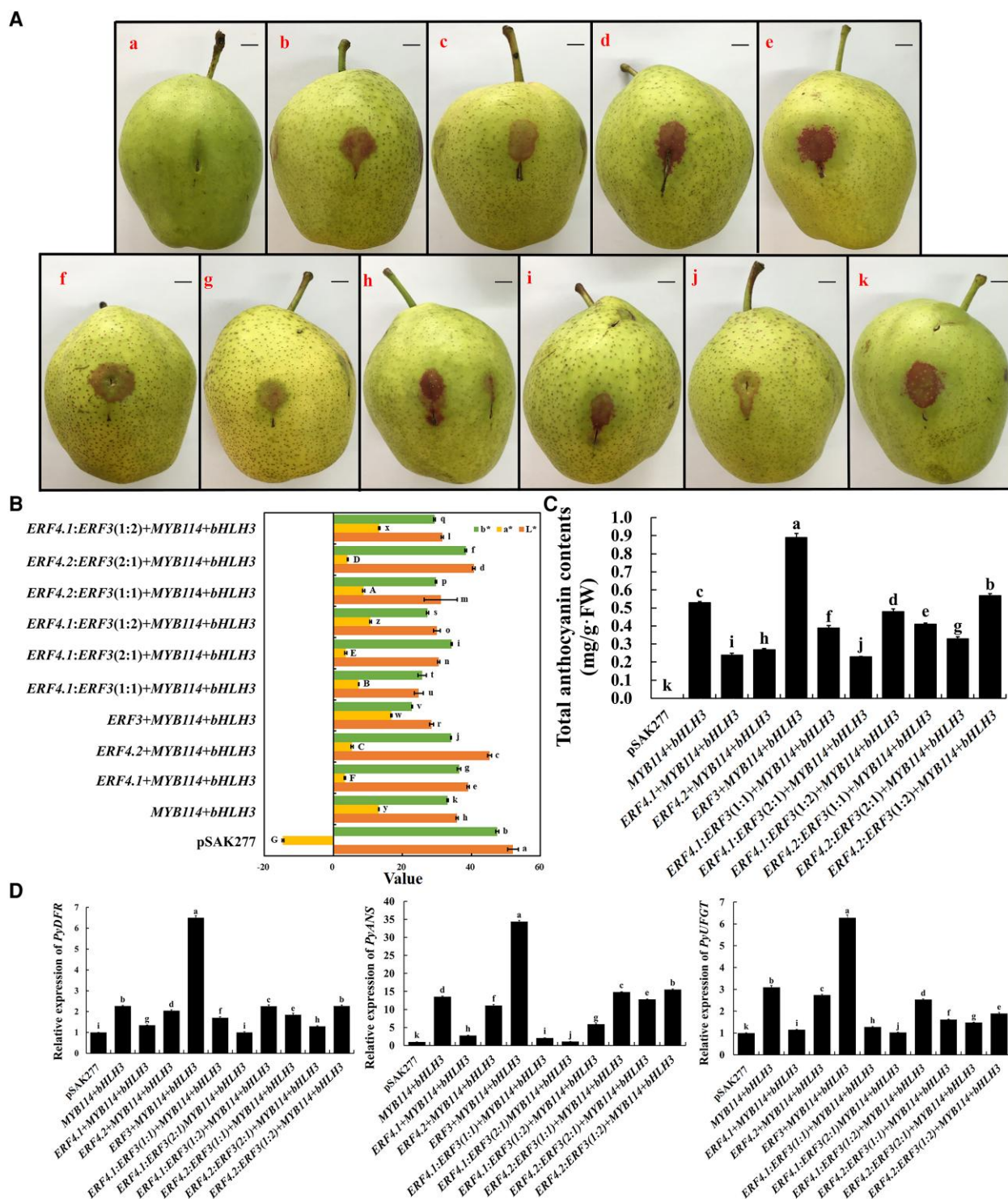


Figure 5. The inhibitory effect of PyERF4.1/PyERF4.2 cotransformation with PyERF3, PyMYB114, and PybHLH3 on anthocyanin biosynthesis by transient expression analysis in pear peels. A) The phenotype of 'Zaosu' pear peels after infiltration: a, pSAK277; b, PyMYB114 + PybHLH3; c, PyERF4.1 + PyMYB114 + PybHLH3; d, PyERF4.2 + PyMYB114 + PybHLH3; e, PyERF3 + PyMYB114 + PybHLH3; f, PyERF4.1:PyERF3 (1:1) + PyMYB114 + PybHLH3; g, PyERF4.1:PyERF3 (2:1) + PyMYB114 + PybHLH3; h, PyERF4.1:PyERF3 (1:2) + PyMYB114 + PybHLH3; i, PyERF4.2:PyERF3 (1:1) + PyMYB114 + PybHLH3; j, PyERF4.2:PyERF3 (2:1) + PyMYB114 + PybHLH3; k, PyERF4.2:PyERF3 (1:2) + PyMYB114 + PybHLH3. Scale bars = 1 cm. B) The difference in color is indicated by the values of L*, a* and b*. L* indicates luminance; a* indicates a range from green to magenta; b* indicates a range from yellow to blue. Bars indicate mean values $\pm$ SD from 6 biological replicates. C) Total anthocyanin contents in transformed pear peels. Bars indicate mean values $\pm$ SD from 3 biological replicates. D) Relative expression of PyDFR, PyANS, and PyUGT. Data are presented as means $\pm$ SD ($n = 3$). Statistical analysis was carried out with one-way ANOVA, and significance was marked with different letters ($P < 0.05$).

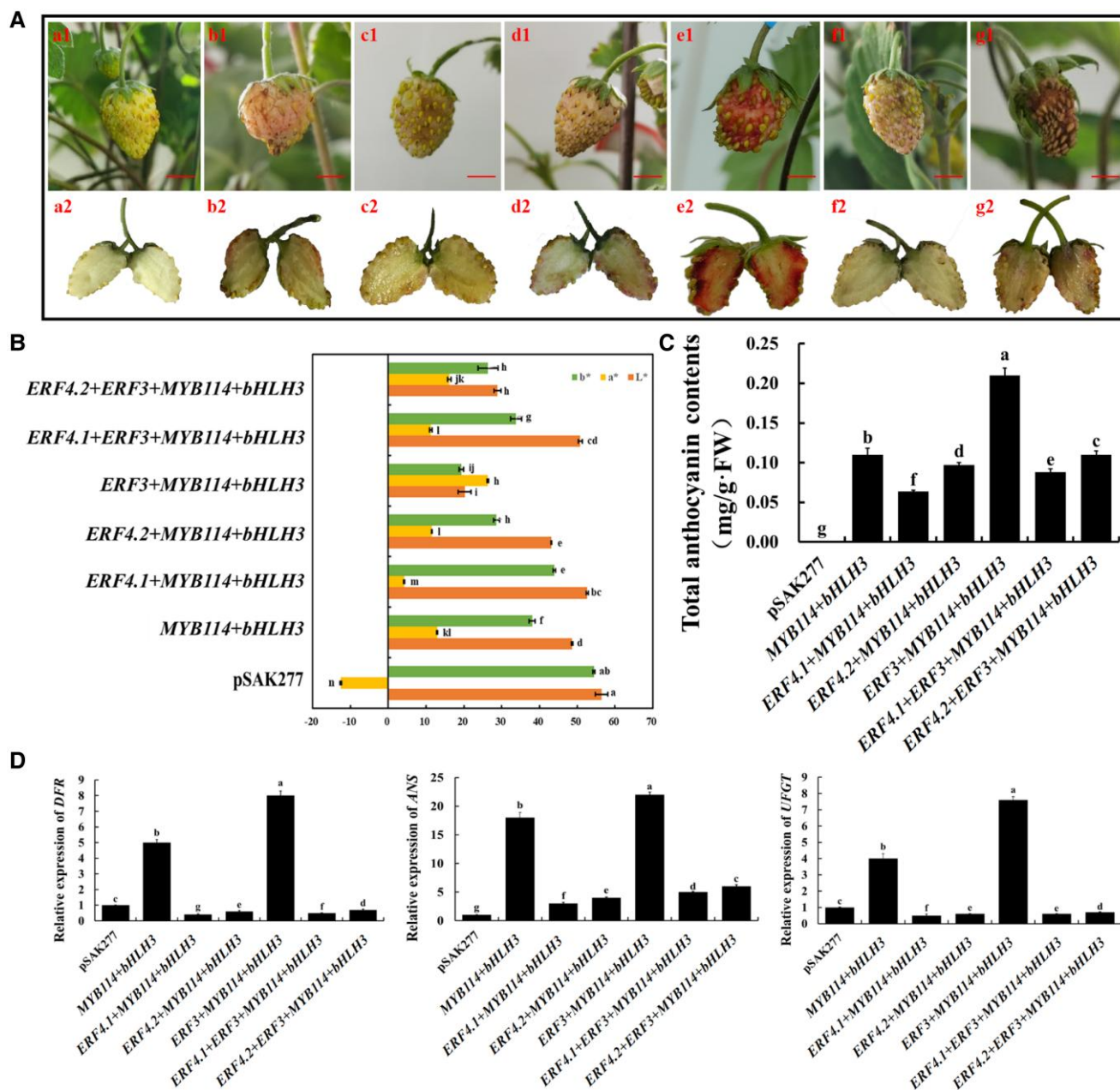


Figure 6. PyERF4.1/PyERF4.2 cotransformation with PyERF3, PyMYB114, and PybHLH3 inhibits anthocyanin biosynthesis by transient expression analysis in strawberry receptacles. A) The phenotype of strawberry receptacles after infiltration: a, pSAK277; b, PyMYB114 + PybHLH3; c, PyERF4.1 + PyMYB114 + PybHLH3; d, PyERF4.2 + PyMYB114 + PybHLH3; e, PyERF3 + PyMYB114 + PybHLH3; f, PyERF4.1 + PyERF3 + PyMYB114 + PybHLH3; g, PyERF4.2 + PyERF3 + PyMYB114 + PybHLH3. 1, for the overall view. 2, for the section view. Scale bars = 5 mm. B) The difference in color is indicated by the values of L^* , a^* , and b^* . L^* indicates luminance; a^* indicates a range from green to magenta; b^* indicates a range from yellow to blue. Bars indicate mean values $\pm$ SD from 6 biological replicates. C) Total anthocyanin contents in transformed strawberry receptacles. Bars indicate mean values $\pm$ SD from 3 biological replicates. D) Relative expression of FvDFR, FvANS, FvUGT. Data are presented as means $\pm$ SD ($n = 3$). Statistical analysis was carried out with one-way ANOVA, and significance was marked with different letters ($P < 0.05$).

(FvDFR, FvANS, and FvUGT) expression levels in strawberry receptacles when cotransformed with PyMYB114–PybHLH3 or PyERF3–PyMYB114–PybHLH3 ($P < 0.05$) (Fig. 6D). Meanwhile, PyERF4.1 had a stronger inhibitory effect than PyERF4.2. Taken together, PyERF4.1/PyERF4.2 inhibits anthocyanin biosynthesis promoted by the PyERF3–PyMYB114–PybHLH3 complex in strawberry receptacles.

PyERF4.1/PyERF4.2 interacts with PyERF3 through the EAR motif

In order to explore the potential interaction between PyERF4.1/PyERF4.2 and PyERF3, PyMYB114, PybHLH3, a *Nicotiana benthamiana*-based firefly luciferase complementation (FLC) assay was performed. PyERF4.1 and PyERF4.2 were inserted into the NLuc, while PyERF3, PyMYB114, and

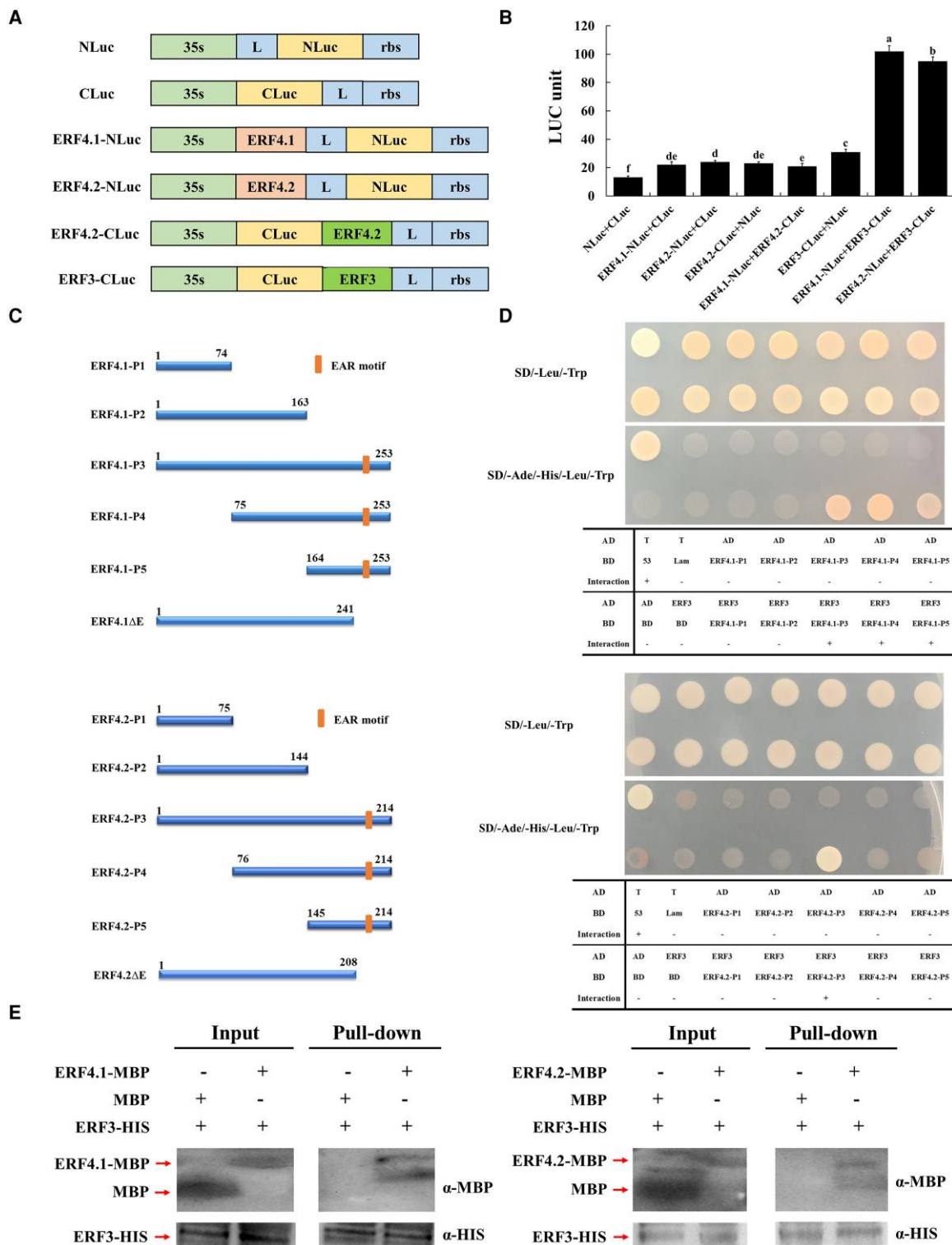


Figure 7. PyERF4.1/PyERF4.2 interacted with PyERF3 through the EAR motif. A) Model of the NLuc, CLuc, and NLuc/CLuc constructs. B) Verification of the interaction between PyERF4.1/PyERF4.2 and PyERF3 via FLC assays. Bars indicate mean values $\pm$ SD from 6 biological replicates. C) Protein fragment of PyERF4.1 and PyERF4.2. P1–P5, part1 to part5, the different amino acid residues of PyERF4.1 and PyERF4.2; EAR motif, ERF-associated-amphiphilic repression motif; Δ E, EAR motif deletion. D) Interaction of PyERF4.1/PyERF4.2 with PyERF3 verified by Y2H assays. E) Validation of PyERF4.1/PyERF4.2 and PyERF3 interactions with pull-down assays. α -MBP, MBP antibody; α -HIS, HIS antibody. F) Interaction of PyERF4.1/PyERF4.2 with PyERF3 through the EAR motif verified by Y2H assays. G) Interaction of PyERF4.1 with PyERF4.2 verified by Y2H assays. H) A luciferase complementation assay showing that PyERF4.1/PyERF4.2 weakened the interaction between PyERF3 and PyMYB114. Bars indicate mean values $\pm$ SD from 6 biological replicates. Statistical analysis was carried out with one-way ANOVA, and significance was marked with different letters ($P < 0.05$).

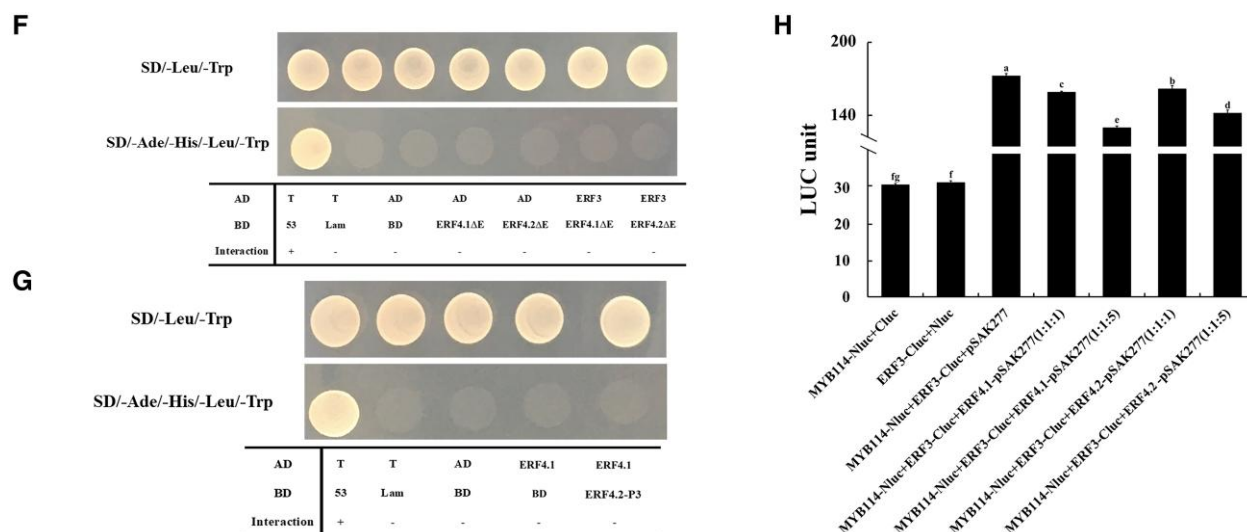


Figure 7. Continued

PybHLH3 were inserted into the CLuc, respectively (Fig. 7A; Supplemental Fig. S4A). Co-expression of PyERF4.1/PyERF4.2-NLuc and PyERF3-CLuc showed strong luciferase enzyme activity. On the contrary, no substantial luciferase enzyme activity was observed for other groups, including PyERF4.1/PyERF4.2-NLuc with CLuc, PyERF4.2/PyERF3-CLuc with NLuc, and PyERF4.1-NLuc with PyERF4.2-CLuc (Fig. 7B). When PyERF4.1/PyERF4.2-NLuc and PyMYB114-CLuc or PybHLH3-CLuc were co-expressed, no substantial luciferase enzyme activity was observed (Supplemental Fig. S4, B and C). Thus, it could be concluded that PyERF4.1/PyERF4.2 showed interactions with PyERF3 and no interactions with PyMYB114 and PybHLH3. Moreover, there was no interaction between PyERF4.1 and PyERF4.2.

Next, the interaction of PyERF4.1/PyERF4.2 with PyERF3, PyMYB114, and PybHLH3 was verified by yeast two-hybrid (Y2H) assays. *PyERF4.1* and *PyERF4.2* were inserted into pGBKT7 as full-length cDNA or as C-/N-terminal deletions (Fig. 7C), and *PyERF3*, *PyMYB114*, and *PybHLH3* full-length cDNAs were inserted into pGADT7. As shown in Fig. 7D, the 5 parts of PyERF4.1/PyERF4.2 (P1–P5) and PyERF3 did not show reporter activity in yeast. When the full-length PyERF4.1^{1–253} (P3) and 2 C-terminal parts PyERF4.1^{75–253} (P4), PyERF4.1^{164–253} (P5) were cotransformed with PyERF3, there was growth on both SD/-Trp/-Leu and SD/-Trp/-Leu/-His/-Ade medium. Thus, the interaction fragment of PyERF4.1 with PyERF3 was located within the C-terminal 90 amino acids (residues 164–253, P5). Meanwhile, when the full-length part PyERF4.2^{1–214} (P3) was cotransformed with PyERF3, there was growth on both SD/-Trp/-Leu and SD/-Trp/-Leu/-His/-Ade medium. Furthermore, none of the 5 parts of PyERF4.1 and PyERF4.2 showed reporter activity when cotransformed into yeast with PyMYB114 or PybHLH3 (Supplemental Fig. S4, D–G). Taken together, these results were consistent with the FLC assay,

suggesting that PyERF4.1/PyERF4.2 have interactions with PyERF3 and no interactions with PyMYB114 and PybHLH3.

A pull-down assay was used to further validate the interaction of PyERF4.1/PyERF4.2 with PyERF3. *PyERF4.1* and *PyERF4.2* were constructed into the pMAL-c2x vector with MBP protein tag, and *PyERF3* was constructed into the pCold-TF vector with HIS protein tag. As shown in Fig. 7E, *PyERF4.1* and *PyERF4.2* proteins with MBP tag could be pulled down by *PyERF3* protein with HIS tag, indicating that there is protein interaction of PyERF4.1/PyERF4.2 with PyERF3, which was consistent with the results of the FLC and Y2H assays.

Y2H assays showed that PyERF4.1 interacts with PyERF3 through the C-terminal 90 amino acids, the full-length of PyERF4.2 interacts with PyERF3, while the EAR motif of PyERF4.1 and PyERF4.2 were located at the C-terminus. Therefore, to verify whether PyERF4.1 and PyERF4.2 interact with PyERF3 through EAR motif, Y2H assay was performed on 2 fragments with deleted EAR motif, named as PyERF4.1ΔE and PyERF4.2ΔE (Fig. 7C). The interaction of PyERF4.1/PyERF4.2 with PyERF3 disappeared after the EAR motif deletion, indicating that PyERF4.1/PyERF4.2 interacts with PyERF3 through EAR motif. Besides, Y2H assay also demonstrated no interaction between PyERF4.1 and PyERF4.2 (Fig. 7, F and G).

To investigate the effect of PyERF4.1/PyERF4.2 on the interaction between PyMYB114 and PyERF3, a luciferase complementation assay was performed. The constructed PyMYB114-Nluc, PyERF3-Cluc, and PyERF4.1/PyERF4.2-pSAK277 recombinant vectors were coinfiltrated into *N. benthamiana* leaves. Co-expressed PyMYB114-Nluc and PyERF3-Cluc showed strong luciferase enzyme activity, but not in the negative controls. The luciferase enzyme activity significantly reduced when PyERF4.1/PyERF4.2-pSAK277

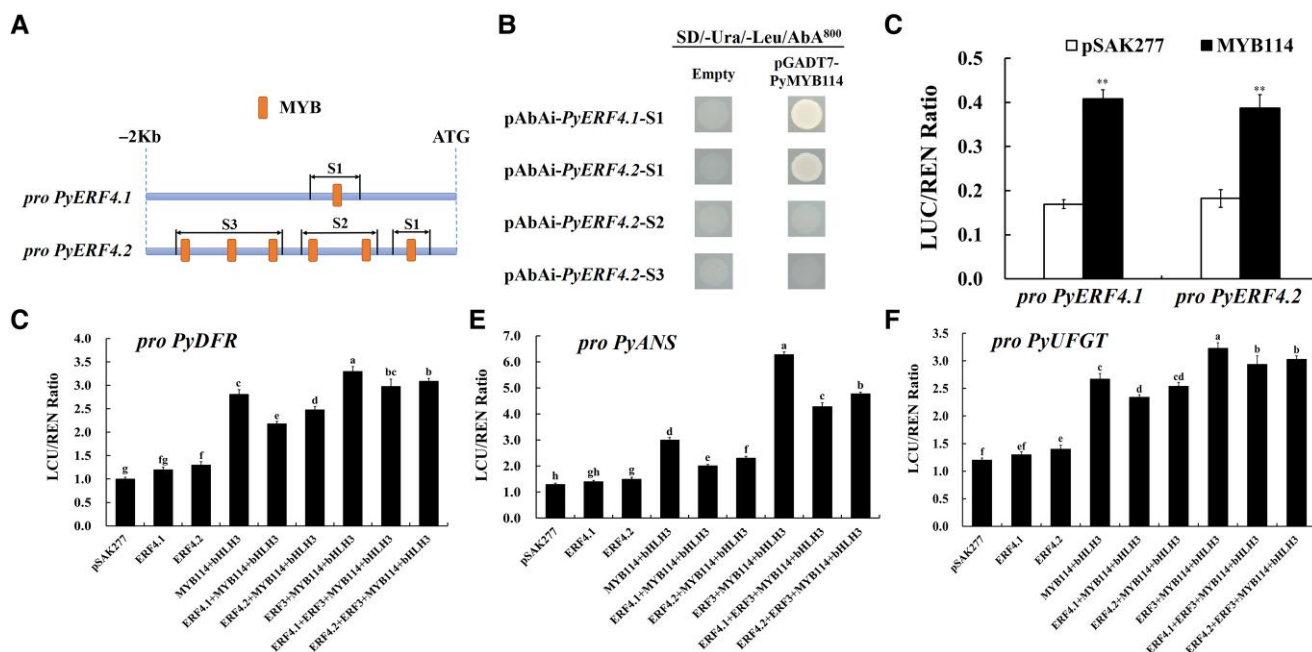


Figure 8. PyMYB114 binds to the *PyERF4.1/PyERF4.2* promoter and regulates transcription of anthocyanin biosynthesis genes. A) Schematic of the *PyERF4.1* and *PyERF4.2* promoters. The cis-acting elements of the promoter region were predicted using the PlantPAN 3.0 database and segmented (S1–S3). Pro, promoter. The MYB is MYBs cis-acting element. B) Y1H assays between PyMYB114 and the *PyERF4.1/PyERF4.2* promoter. The yeast colonies were screened on SD/–Ura/–Leu/AbA⁸⁰⁰ plates. C) The dual-luciferase reporter assay verifies PyMYB114 binding to *PyERF4.1* and *PyERF4.2* promoters. Data are presented as means $\pm$ SD ($n = 6$). D–F) *PyERF4.1/PyERF4.2* cotransformation with *PyERF3*, *PyMYB114* and *PybHLH3* on the transcription of *PyDFR*, *PyANS*, and *PyUGT* verified by the dual-luciferase reporter assay. Data are presented as means $\pm$ SD ($n = 6$). Statistical analysis was carried out with one-way ANOVA and Student's *t*-test, and significance was marked with asterisks (* $P < 0.05$) or (** $P < 0.01$) or different letters ($P < 0.05$).

was added ($P < 0.05$), and the addition of *PyERF4.1/PyERF4.2*-pSAK277 was inversely associated with the luciferase enzyme activity (Fig. 7H). In summary, it could be concluded that the interaction between *PyERF4.1/PyERF4.2* and *PyERF3* interferes with the binding of *PyERF3* to *PyMYB114*, which in turn affects the stability of *PyERF3*–*PyMYB114*–*PybHLH3* complex.

PyMYB114 activates the *PyERF4.1/PyERF4.2* transcription and regulates the expression of anthocyanin biosynthesis genes in pears

Based on the results of transient expression systems in pear peels and strawberry receptacles, it was found that *PyERF4.1/PyERF4.2* cotransformation with *PyMYB114*–*PybHLH3* inhibited pigment deposition, but neither *PyERF4.1* nor *PyERF4.2* interacted with *PyMYB114* and *PybHLH3*. Furthermore, *PyERF4.1* and *PyERF4.2* expression levels were significantly elevated when *PyMYB114* and *PybHLH3* were transiently overexpressed ($P < 0.01$) (Supplemental Fig. S5A), and *PybHLH3* could not activate the *PyERF4.1* and *PyERF4.2* transcription (Supplemental Fig. S5B). Therefore, it was hypothesized that *PyMYB114* binds to the *PyERF4.1* and *PyERF4.2* promoters, and this regulatory module was verified with yeast one-hybrid (Y1H) assay. Cis-acting elements on the promoter sequence of *PyERF4.1*

and *PyERF4.2* were predicted by PlantPAN 3.0, and the results are presented in Fig. 8A. The upstream 2 kb promoter region of the *PyERF4.1* and *PyERF4.2* was found to contain one or more cis-acting elements of MYB TFs. Then, the promoter segment baits were integrated with the prey vectors pGADT7-*PyMYB114* and introduced into the Y1HGOLD yeast strain, and the results showed that *PyMYB114* could bind to the S1 fragment of *PyERF4.1/PyERF4.2* (Fig. 8B). The trans-activity of *PyMYB114* to the *PyERF4.1/PyERF4.2* was further verified by a dual-luciferase reporter assay in *N. benthamiana* leaves. In Fig. 8C, the transformation of *PyMYB114* had a significant activation effect on the transcription of *PyERF4.1* and *PyERF4.2* compared to that of the empty vector pSAK277 ($P < 0.01$). In conclusion, these results indicated that *PyMYB114* binds to the *PyERF4.1* and *PyERF4.2* promoters, thereby activating their transcription.

PyERF4.1/PyERF4.2 interacted with *PyERF3*, *PyMYB114*, and *PybHLH3* to inhibit anthocyanin biosynthesis, but the downstream target genes on which they act remain to be determined. Therefore, the dual-luciferase reporter assay in *N. benthamiana* leaves was performed to investigate the effect of *PyERF4.1/PyERF4.2* cotransformation with *PyERF3*, *PyMYB114*, and *PybHLH3* on the downstream target genes *PyDFR*, *PyANS*, and *PyUGT*, which are associated with anthocyanin biosynthesis. In contrast to the empty vector pSAK277 transformation, *PyERF3*,

PyMYB114, and PybHLH3 cotransformation was found to significantly activate PyDFR, PyANS, and PyUGT promoters, whereas the activation was significantly attenuated when cotransformed with PyERF4.1/PyERF4.2 ($P < 0.05$) (Fig. 8, D–F). Also, the de-activating effect of PyERF4.1 on PyANS promoters was significantly stronger than that of PyERF4.2 ($P < 0.05$). Therefore, PyERF4.1/PyERF4.2 damaged the activating effect of the complex PyERF3–PyMYB114–PybHLH3 on the promoters of anthocyanin biosynthesis genes.

EAR motif deletion or mutation diminishes the inhibitory effect of PyERF4.1 and PyERF4.2 on anthocyanin biosynthesis in red-skinned pears

Although PyERF4.1 and PyERF4.2 could inhibit anthocyanin biosynthesis in pears, PyERF4.1 showed stronger inhibition, presumably because it had a complete EAR motif, whereas PyERF4.2 had an amino acid mutation in its EAR motif. To verify this hypothesis, the EAR motifs of PyERF4.1/PyERF4.2 were deleted (PyERF4.1ΔE/PyERF4.2ΔE), and the Met residue was mutated to the Leu residue in the EAR motif of PyERF4.2 (PyERF4.2M) to mimic an intact EAR motif, and the vectors were cotransformed with PyERF3, PyMYB114, and PybHLH3 into ‘Zaosu’ pears (Supplemental Fig. S6). When PyERF4.1ΔE or PyERF4.2ΔE were cotransformed with PyERF3–PyMYB114–PybHLH3, the same substantial pigment deposition occurred as when PyERF3–PyMYB114–PybHLH3 were cotransformed. In contrast, when PyERF4.2M was cotransformed with PyERF3–PyMYB114–PybHLH3, less pigmentation appeared than when cotransformed with PyERF4.2 (Fig. 9A). The color changes and anthocyanin contents of the injection sites were consistent with the phenotypes (Fig. 9, B and C). RT-qPCR analysis showed that the PyDFR, PyANS, and PyUGT expression level were significantly higher when PyERF3–PyMYB114–PybHLH3 were cotransformed or when PyERF4.1ΔE/PyERF4.2ΔE was cotransformed ($P < 0.05$), while when PyERF3–PyMYB114–PybHLH3 were cotransformed with PyERF4.1/PyERF4.2 or PyERF4.2M, they were significantly reduced ($P < 0.05$) (Fig. 9D). In summary, it could be concluded that EAR motif deletion diminishes the inhibitory effect of PyERF4.1 and PyERF4.2 on anthocyanin biosynthesis, whereas a mutation of the EAR motif in PyERF4.2 to intact EAR motif (LxLxM to LxLxL) strengthened the inhibitory effect on anthocyanin biosynthesis in red-skinned pears.

Discussion

PyERF4.1 and PyERF4.2 inhibit anthocyanin biosynthesis in red-skinned pears

Recently, red-skinned pear is popular because of rich anthocyanins, but its coloring is unstable during the development process, thus the molecular mechanism of anthocyanin biosynthesis in many species has been widely concerned and studied (Li et al. 2020). Many AP2/ERF TFs have been

reported to positively regulate anthocyanin biosynthesis in plants, such as AtERF4 and AtERF8 in *Arabidopsis* (*Arabidopsis thaliana*) (Koyama and Sato 2018), MdERF1B (Zhang et al. 2018), MdERF3 (An et al. 2018), and MdERF38 (An et al. 2020b) in apple. In red-skinned pear, PyERF3 (Yao et al. 2017), Py4ERF24, Py12ERF96 (Ni et al. 2019), PyERF22 (Wu et al. 2020), and PyERF105 (Ni et al. 2021) all promote anthocyanin biosynthesis. The negative regulation of pigment metabolism by ERFs is equally important, but little attention has been paid to it. Most of the available reports are about the regulation of the color change in fruit ripening process, such as tomato SlERF6 and SlERF.F12 (Lee et al. 2012; Deng et al. 2022), loquat (*Eriobotrya japonica*) EjERF11 (Zeng et al. 2015), banana (*Musa nana* Lour.) MaERF11 (Han et al. 2016), and apple (*Malus domestica*) MdERF4 (Hu et al. 2022).

In this study, expression of 2 AP2/ERFs, PyERF4.1 and PyERF4.2 screened from the transcriptome data of ‘Starkrimson’ pear and its green mutant during 3 fruit developmental periods was substantially and negatively correlated with anthocyanin contents as well as PyERF3, PyMYB114, PybHLH3, PyDFR, PyANS, and PyUGT expression by bioinformatics and correlation analysis (Figs. 1 and 2). PyERF4.1 and PyERF4.2 as repressors regulated anthocyanin biosynthesis in overexpression pear calli and stable hereditary tomato fruits (Figs. 3 and 4). Further studies indicated that PyERF4.1 and PyERF4.2 inhibited anthocyanin biosynthesis with PyERF3–PyMYB114–PybHLH3 complex in strawberry receptacles and pears as verified by transient expression and RT-qPCR analysis (Figs. 5 and 6). Pigment deposition is a phenotypic feature during fruit ripening. The present study verified that PyERF4.1 and PyERF4.2 inhibited anthocyanin biosynthesis in red-skinned pear fruit, which provides a more in-depth study for the molecular mechanism of ERFs negatively regulating the color change in fruit-ripening process. In summary, we provided direct evidence highlighting the involvement of PyERF4.1 and PyERF4.2 as repressors of anthocyanin biosynthesis in red-skinned pears.

The inhibitory effect of AP2/ERFs on anthocyanin biosynthesis is dependent on EAR motif in pears

EAR motif-mediated repression of transcription is the predominant form of repression in plants (Kagale et al. 2010; Choi et al. 2018), and an increasing amount of studies have reported that deletion and mutation of EAR motif affect the function of genes, e.g. a single amino acid mutant in the EAR motif of IbMYB44.2 reduced the inhibition of anthocyanin accumulation in the purple-fleshed sweet potato (*Ipomoea batatas* L.) (Li et al. 2021); EAR motif mutation (C-G) in MdERF4 reduces the expression of MdERF3, and promoted apple fruit ripening (Hu et al. 2022). In this study, the PyERF4.1 amino acid sequence had an intact EAR motif at the C-terminus (DLNLxP), while the PyERF4.2 amino acid sequence had an incomplete EAR motif with a mutated amino acid (LxLxM) (Fig. 1C). Moreover, PyERF4.1 was verified to have a stronger inhibitory effect than PyERF4.2 in strawberry

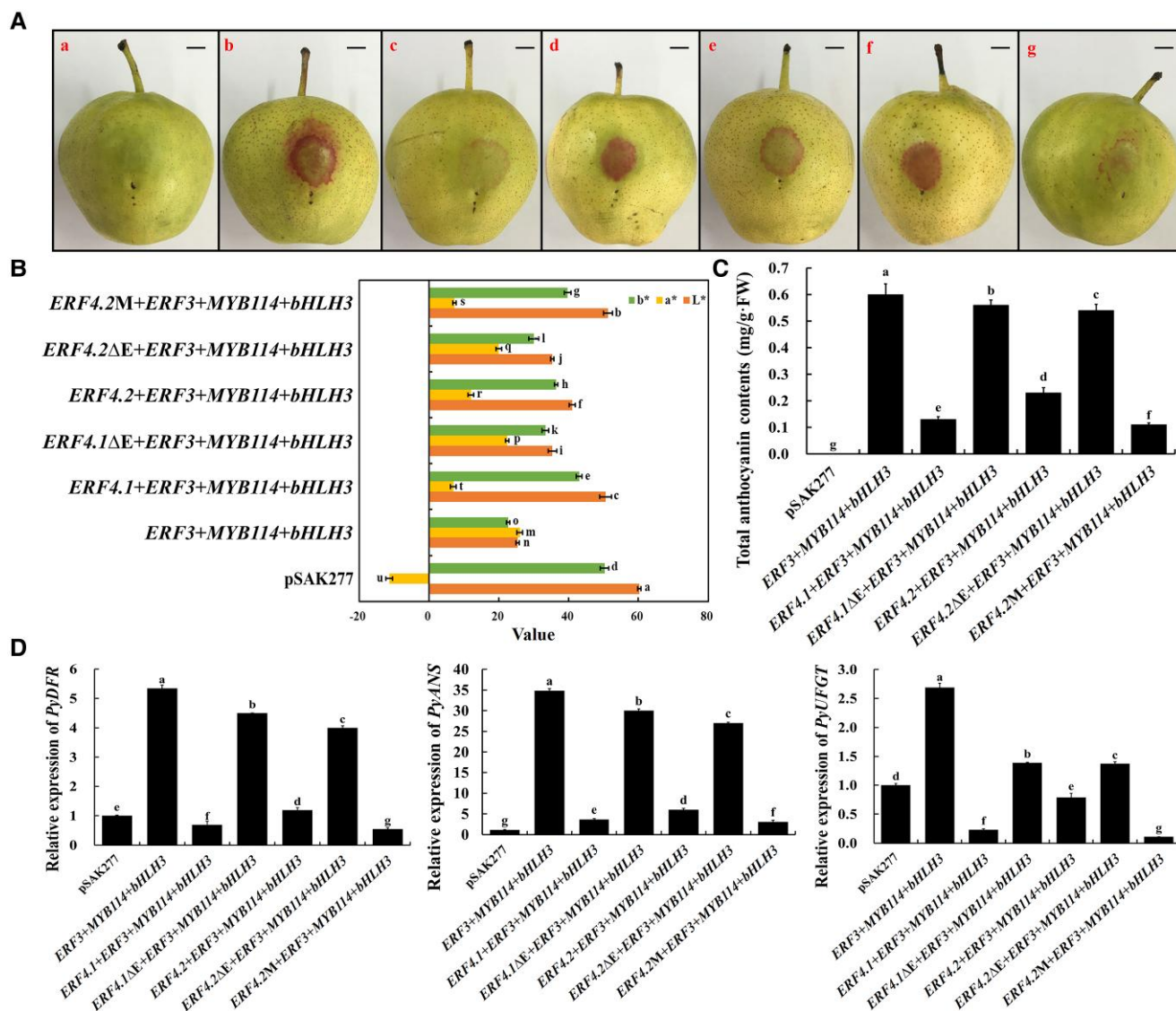


Figure 9. Effect of EAR motif integrity on the function of PyERF4.1 and PyERF4.2 in inhibiting anthocyanin biosynthesis. A) The phenotype of ‘Zaosu’ pear peels after infiltration: a, pSAK277; b, PyERF3 + PyMYB114 + PybHLH3; c, PyERF4.1 + PyERF3 + PyMYB114 + PybHLH3; d, PyERF4.1ΔE + PyERF3 + PyMYB114 + PybHLH3; e, PyERF4.2 + PyERF3 + PyMYB114 + PybHLH3; f, PyERF4.2ΔE + PyERF3 + PyMYB114 + PybHLH3; g, PyERF4.2 M + PyERF3 + PyMYB114 + PybHLH3. ΔE, EAR motif deletion. M, EAR motif mutation. Scale bars = 1 cm. B) The difference in color is indicated by the values of L^* , a^* , and b^* . L^* indicates luminance; a^* indicates a range from green to magenta; b^* indicates a range from yellow to blue. Bars indicate mean values $\pm$ SD from 6 biological replicates. C) Total anthocyanin contents in transformed pear peels. Bars indicate mean values $\pm$ SD from 3 biological replicates. D) Relative expression of *PyDFR*, *PyANS*, and *PyUGT*. Data are presented as means $\pm$ SD ($n = 3$). Statistical analysis was carried out with one-way ANOVA, and significance was marked with different letters ($P < 0.05$).

receptacles, pear fruits, and transgenic pear calli (Figs. 3, 5, and 6), presumably due to the incomplete EAR motif of PyERF4.2. This is similar to previous studies reporting that a single amino acid mutation in the EAR motif diminishes the inhibitory effect. Some studies reported that the deletion or mutation of the EAR motif abolished the cell death-inducing ability of StERF3 in potato (*Solanum tuberosum*) and the inhibitory effect of SlERF.F12 on ripening in tomato (Deng et al. 2022; Qi et al. 2022). Here, transient expression assays in pears of PyERF4.1 and PyERF4.2 with EAR motif

deletions showed that EAR motif deletion abolished the inhibitory effect of PyERF4.1/PyERF4.2 on anthocyanin biosynthesis, whereas the inhibitory effect of PyERF4.2 was enhanced after incomplete EAR motif mutated to an intact form (Fig. 9). These results demonstrate that EAR motif deletion diminishes the inhibitory effect of PyERF4.1/PyERF4.2 on anthocyanin biosynthesis in pears. Consistently, previous studies reported that EAR motif deletion or mutation promotes ripening in tomato and apple by Deng et al. (2022) and Hu et al. (2022).

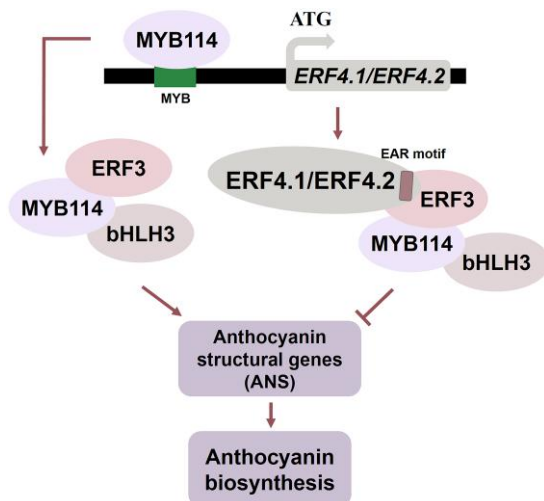


Figure 10. A regulatory model for PyERF4.1 and PyERF4.2 inhibition of anthocyanin biosynthesis in red-skinned pears.

A regulatory model of a feedback regulatory loop balancing anthocyanin biosynthesis in red-skinned pears

The MBW complex is a major regulator in the plant anthocyanin biosynthetic pathway and has been widely reported in tomato (Yan et al. 2020), apple (An et al. 2020a), and pear (Cui et al. 2021). An increasing number of studies have reported that AP2/ERFs regulate anthocyanin biosynthesis in pears through the MBW complex (Ni et al. 2019, 2021; Wu et al. 2020). This study verified that PyMYB114 binding to the promoter of *PyERF4.1/PyERF4.2*, activating their transcription and leading to a decrease in transcriptional activity of anthocyanin biosynthesis key gene *PyANS*, thereby inhibiting red-skinned pear fruit coloration (Fig. 8). Previous studies reported that MdMYB1 activates the promoter of transcriptional activator *MdERF3* to promote ethylene production in apples (An et al. 2018). In the present study, PyMYB114 activates the promoter of transcriptional repressor *PyERF4.1/PyERF4.2* to inhibit anthocyanin biosynthesis in red-skinned pears. This is similar to previous studies, while MYBs activate the promoter of repressor which provides a distinct module for the mechanism of red-skinned pear coloration regulation.

It has been shown that PyERF3 interacts with PyMYB114 and forms the PyERF3–PyMYB114–PybHLH3 complex to activate downstream target genes and enhance anthocyanin biosynthesis in red-skinned pears (Yao et al. 2017). In this study, PyERF4.1 and PyERF4.2 were found to interact with PyERF3 through EAR motif to affect the stability of the PyERF3–PyMYB114–PybHLH3 complex and thus inhibit red-skinned pears anthocyanin biosynthesis (Fig. 7). This model of regulation has similarities with the interaction of MdERF2 and MdERF3 proteins that inhibits the association of MdERF3 to the *MdACS1* promoter, thereby negatively affecting ethylene biosynthesis and fruit ripening in apples, as

previously reported by Li et al. (2016). Furthermore, we found that PyMYB114 could not only promote anthocyanin accumulation via forming a complex with PybHLH3–PyERF3, but also activate the repressors *PyERF4.1* and *PyERF4.2* to affect the stability of PyERF3–PyMYB114–PybHLH3 complex, leading to a decrease in the accumulation of anthocyanin in pears. Thus, a feedback regulation loop was formed to balance the excessive accumulation of anthocyanins in red-skinned pears. In summary, a regulatory model of *PyERF4.1/PyERF4.2* inhibition of anthocyanin biosynthesis in red-skinned pears can be summarized: PyMYB114 binds to the promoter of *PyERF4.1/PyERF4.2* and activates its transcription in order to balance the accumulation of anthocyanin in red-skinned pears, then *PyERF4.1/PyERF4.2* interact with PyERF3 through the EAR motif to break the stability of the PyERF3–PyMYB114–PybHLH3 complex, which inhibited anthocyanin biosynthesis gene *PyANS* transcription, resulting in the decreased anthocyanin in red-skinned pears (Fig. 10).

Materials and methods

Plant materials and growth conditions

The red-skinned pears ‘Hongzaosu’ (*P. pyrifolia* Nakai) and the green-skinned pears ‘Zaosu’ (*P. pyrifolia* Nakai) used in this study were provided by Shandong Institute of Pomology (Tai’an, China). The fruits were harvested at 30, 60, 90 DAFB, photographed and sampled for peels, frozen in liquid nitrogen and stored at -80°C . The ‘Zaosu’ pears at ripening stage and diploid strawberry (*F. vesca*) ‘Yellow Wonder’ 5AF7 at 2 weeks after flowering grown in a greenhouse under artificial light (16 h at 25°C in the day and 8 h at 18°C in the night), were used for transient expression assays. The 4-week-old tobacco (*N. benthamiana*) leaves were used for dual-luciferase reporter system and FLC assays. The phenotype of pear peels, tobacco leaves, and strawberry receptacles were observed 6 d postinjection, and then the samples were frozen in liquid nitrogen and stored at -80°C .

Tomato (*S. lycopersicum*) cv. Micro-Tom was selected as the WT in this study. The WT, overexpression and deletion mutation tomato plants were grown in a greenhouse under artificial light (16 h at 25°C in the day and 8 h at 18°C in the night). Red ripening tomato fruits of stable lines with T2 generation overexpression (*ERF4.1-OE*) and homozygous mutants (*erf4.1*) were sampled and frozen in liquid nitrogen and stored at -80°C . Each line sample contained at least 6 fruits.

Bioinformatic analysis

The 114 differentially expressed genes AP2/ERFs from transcriptome data of red-skinned pear ‘Starkrimson’ (*P. communis* L.) and its green mutant reported by Yang et al. (2015) were analyzed. Heat map was established with the R script, and the phylogenetic tree was constructed by the MEGA7 Program according to the neighbor-joining method, bootstrap analysis (1,000 replicates). The amino acid sequence

alignment analysis was performed through the DNAMAN Program.

RT-qPCR analysis

Total RNA was extracted from pear peel, strawberry receptacle, pear calli, and tomato fruit samples by the Plant RNA Isolation Kit (Foregene, Chengdu, China), and first-strand RNA was synthesized using a Primer Reverse Transcription Master Mix Kit (Takara, Tokyo, Japan). Primers for RT-qPCR are listed in [Supplemental Table S1](#). *PyTubulin*, *PyActin*, *Fv26S*, *FvActin*, *SlTubulin*, and *SlActin* expression levels were used as normalized genes, and relative gene expression was determined by the $2^{-\Delta\Delta CT}$ method (Pirrello et al. 2006). Three biological replicates were used for all analyses.

Genes cloning and recombinant vectors construction

Full-length sequences of *PyERF4.1* and *PyERF4.2* genes cloned from the cDNA of ‘Starkrimson’ pears. For the construction of stable genetic lines, full-length coding sequences of *PyERF4.1* and *PyERF4.2* were inserted into the pCambia1300-221 vector at the restriction sites of *Bam*HI and *Sac*I under the control of 35S promoter. For the transient transformed expression analysis, full-length coding sequences of *PyERF4.1*, *PyERF4.2*, *PyERF3*, *PyMYB114*, and *PybHLH3*, and the sequences of *PyERF4.1ΔE* (bases 1–723 bp) and *PyERF4.2ΔE* (bases 1–624 bp) with deleted EAR motif as well as *PyERF4.2M* (methionine mutated to leucine) with mutated EAR motif were inserted into the pSAK277 vector at the restriction sites of *Eco*R I and *Xba*I under the control of 35S promoter. For RNA interference (RNAi)-induced gene expression silencing, a 300 bp *PyERF4.1*-specific DNA fragment (bases 374–673 bp) and a 279 bp *PyERF4.2*-specific DNA fragment (bases 364–642 bp) were inserted into the pSAK277 vector, respectively. Primers used to construct the recombinant plasmids are listed in [Supplemental Table S2](#).

Transient transformation and RNAi-induced silencing in pear fruits and heterologous expression in strawberry receptacles

For the transient transformed expression and RNAi-induced silencing analysis, the recombinant plasmids constructed on the pSAK277 vector were transformed into GV3101 strains of *Agrobacterium tumefaciens*, and then *Agrobacterium* cells containing recombinant pSAK277 vector were resuspended in the injection solution containing 10 mmol 2-(*N*-morpholino)ethanesulfonic acid hydrate (MES), 10 mmol $MgCl_2$ and 0.2 μ mol acetosyringone ($OD_{600} = 1.0$), then cultured at 25 °C under 70 rpm for 4 h before injection. The empty vector (pSAK277) was used as a negative control. The detail of the infiltration experiment was carried out following the description by Yao et al. (2017). Strawberry receptacles and pear fruits were incubated in the dark for 24 h after injection, then moved to artificial light (16 h of daylight), and

incubated at 24 °C. After 7 d, pear peels and strawberry receptacles were collected to determine anthocyanin content and for total RNA extraction.

Generation of transgenic pear calli

The induced calli from the flesh of ‘Clapp’s Favorite’ (*P. communis* L.) were subcultured several times, and the rapidly growing soft pear calli was selected and played under dark condition on the Murashige and Skoog (MS) solid medium supplemented with 30 g/L sucrose, 0.5 mg/L 6-benzylaminopurine, and 1.0 mg/L 2,4-dichlorophenoxyacetic acid. Pear calli was transformed as previously reported (Bai et al. 2019). Pear calli were immersed in *A. tumefaciens* strain EHA105 containing pCambia1300-221-*PyERF4.1* or pCambia1300-221-*PyERF4.2* recombinant plasmids for 30 min with the empty vector pCambia1300-221 as a negative control (WT). After 2 d of co-culture, the pear calli were cultured in darkness on the MS solid medium containing appropriate antibiotics for 1 mo at 24 °C and subcultured every 2 weeks. For the light treatment, fresh subcultured calli were treated with continuous light and observed after 7 d.

Generation of stable overexpressed and deletion mutant tomato fruits

The constructed pCambia1300-221-*PyERF4.1* recombinant plasmids was transformed into Micro-Tom via *A. tumefaciens* EHA105-mediated transformation. Stable lines of T2 generation of overexpression (*ERF4.1-OE*) was used for further analysis.

Due to the difficulties of stable pear transformation, homologous genes in tomato were searched using the BLAST program in NCBI with the coding sequence of *PyERF4.1* as a query (<https://www.ncbi.nlm.nih.gov/>). It was found that *SlERF4.1* (Solyc10g009110.1.1) showed high similarity to *PyERF4.1* (41.11% identity at the amino acid level). The pYLCRISPR/Cas9-DN binary vector for plant CRISPR/Cas9-mediated genome editing was gifted by Prof. Yaoguang Liu (South China Agricultural University). Two target sequences of *SlERF4.1* were designed using the CRISPR direct online tool (<http://crispr.dbcls.jp/>). The double-stranded DNA of target sequences was amplified by PCR and cloned into the pYLCRISPR/Cas9-DN binary vector using the Golden Gate ligation method (Engler and Marillonnet 2013). Primers used to construct the recombinant plasmids are listed in [Supplemental Table S2](#).

The constructed pYLCRISPR/Cas9-DN-*SlERF4.1* recombinant plasmids were transformed into Micro-Tom via *A. tumefaciens* EHA105-mediated transformation. Stable lines of T2 generation of homozygous mutants (*erf4.1*) were used for further analysis.

Extraction and determination of anthocyanin contents

The differences in color due to anthocyanins were indicated by L^* , a^* , and b^* values determined by the colorimeter

(WSC-100, Konica Minolta, Tokyo, Japan). The method described by Lee and Wicker (1991) was used to extract and determine the anthocyanin content. Briefly, 0.2 g of sample (pear peel, strawberry receptacle, pear calli, or tomato fruit) was homogenized with 1 mL of 1% (v/v) hydrochloric acid methanol solution. The absorbance was measured at 530, 620 and 650 nm by the Multiskan Spectrum (Thermo Scientific Multiskan GO 1510, Finland), and used the formula $OD = (A_{530} - A_{620}) - 0.1 \times (A_{650} - A_{620})$ to calculate total anthocyanin contents of each sample and expressed as mg/g fresh weight.

Dual-luciferase reporter assay

For the construction of dual-luciferase reporter vector, the 2 kb promoter regions (from the ATG start codon) upstream of *PyERF4.1*, *PyERF4.2*, *PyDFR*, *PyANS*, and *PyUFGT* were cloned with the primers listed in Supplemental Table S2 and inserted into the pGreen II 0800-LUC binary vector. Furthermore, *Agrobacterium* transformation and injection were performed in the same way as for the pear fruit and strawberry transient transformation assays. *Agrobacterium* cells containing the pGreen II 0800-LUC and the pSAK277 recombinant plasmids were mixed in a 1:9 ratio. The *Agrobacterium* cells mixture was injected into the 4-week-old *N. benthamiana* leaves for transient transformation expression assay. The ratio of firefly luciferase (LUC)/*Renilla* luciferase (Ren) was determined by the Dual-Luciferase Reporter Assay System (E1910, Promega, USA).

Firefly luciferase complementation assay

Firefly luciferase complementation (FLC) assay was performed as reported by Chen et al. (2008). Full-length coding sequences of *PyERF4.1* and *PyERF4.2* without the stop codons were cloned into the pCAMBIA1300-NLuc binary vector, and full-length sequences of *PyERF3* and *PyMYB114* were inserted into the pCAMBIA1300-CLuc binary vector. The primer sequences are listed in Supplemental Table S2. *Agrobacterium* transformation and injection were performed as described previously for transient transformation assays in *N. benthamiana* leaves. The firefly luciferase activity was measured after 72 h using the Steady-Glo Luciferase Assay System (E2510, Promega, USA).

Y1H assay

Y1H assay was carried out with the Matchmaker Gold Yeast One-Hybrid System (630491, Clontech, Japan). One segment of the *PyERF4.1* promoter (−905 bp to −558 bp) and 3 segments of the *PyERF4.2* promoter (−2,000 bp to −1,887 bp, −1,640 bp to −1,281 bp, and −1,632 bp to −1,087 bp) were cloned into the pAbAi vector containing *Hind* III and *Xho* I, and the full sequence of *PyMYB114* was cloned into the pGADT7 vector containing *Eco*RI and *Xho* I. The primer sequences used for vector construction are listed in Supplemental Table S2. Then, the prey vectors were transformed into Y1H Gold cells containing the pAbAi-bait and detected on SD/-Ura/-Leu/AbA plates.

Y2H assay

Y2H assay was carried out with the Matchmaker Gold Yeast Two-Hybrid System. Six fragments of *PyERF4.1* gene (*ERF4.1*^{1–74}, *ERF4.1*^{1–163}, *ERF4.1*^{1–253}, *ERF4.1*^{75–253}, *ERF4.1*^{164–253}, and *ERF4.1*^{1–241}), and 6 fragments of *PyERF4.2* gene (*ERF4.2*^{1–75}, *ERF4.2*^{1–144}, *ERF4.2*^{1–214}, *ERF4.2*^{76–214}, *ERF4.2*^{145–214}, and *ERF4.2*^{1–208}) were inserted into the pGBKT7 vector containing *Nde* I and *Pst* I, and the full sequences of *PyERF4.1*, *PyERF3*, and *PyMYB114* were cloned into pGADT7 at the restriction sites of *Eco*RI and *Xho* I. The primer sequences used for vector construction are listed in Supplemental Table S2. Then, the recombinant plasmids were cotransformed into Y2H Gold cells by LiCl-PEG method, and the interactions were detected on SD/-Leu/-Trp/-His/-Ade plates. The pGADT7-T and pGBKT7-53 or pGADT7-T and pGBKT7-Lam were cotransformed as positive and negative controls, respectively.

Pull-down assay

Full-length coding sequences of *PyERF4.1* and *PyERF4.2* were cloned into pMAL-p5x and pMAL-c2x vectors with *Bam*H I and *Hind* III, and the full-length coding sequence of *PyERF3* was cloned into pCold TF DNA vector with *Bam*H I and *Xba* I. The primer sequences used for vector construction are shown in Supplemental Table S2. Next, *PyERF4.1*-MBP, *PyERF4.2*-MBP, and *PyERF3*-HIS fusion proteins were expressed in DE3 *Escherichia coli* cells, and *PyERF3*-HIS was purified using Ni Sepharose 6 Fast Flow (17-5318-06, GE Healthcare, Sweden). The collected elution extract was blotted with MBP or HIS antibodies (Beyotime, Shanghai, China).

Accession numbers

Sequence data for this article is available in the genome database for the Rosaceae (<http://www.rosaceae.org>), the National Center for Biotechnology Information (NCBI, <https://www.ncbi.nlm.nih.gov/>), or the Solanaceae (<https://solgenomics.net/>) under the following accession numbers: *PyERF4.1* (ON652752), *PyERF4.2* (ON652753), *SIAP2a* (NP_001234452.1), *SIERF6* (NP_001266125.1), *MdERF1* (BAF434.19.1), *MdERF2* (NP_001280848.1), *AdERF9* (ADJ67438.1), *MaERF11* (XP_009412068), *MdERF4* (NP_001306183.1), *EjERF11* (AKN10304.1), *PyTubulin* (XM_009376045.2), *PyActin* (JN684184), *PyERF3* (ASY06613.1), *PyMYB114* (ASY06612.1), *PybHLH3* (XP_048442703.1), *PyANS* (Pbr001543.2), *PyDFR* (Pbr020145.1), *PyUFGT* (Pbr039986.1), *Fv26S* (gene11892), *FvActin* (gene22626), *FvANS* (gene32347), *FvDFR* (gene15174), *FvUFGT* (gene12591), *SlTubulin* (Solyc08g006890), *SlActin* (Solyc03g078400), *SIERF4.1* (Solyc10g009110.1.1), *SIERF3* (XP_004249668.1), *SIMYB114* (XP_004237817.1), *SIANS* (Solyc08g080040), *SlDFR* (Solyc02g085020), *SlUFGT* (XM_004247965).

Acknowledgments

We thank Dr Andrew C. Allan, Dr Lin-Wang Kui and Dr Richard Espley for the dual vector pGreen II 0800-LUC in

The New Zealand Institute for Plant & Food Research Limited, Auckland, New Zealand.

Author contributions

H.Y.S., G.F.Y., and H.Z. conceived and designed the experiments; H.Y.S. and G.F.Y. performed the experiments; S.W.W. provided the experiment materials; H.Y.S. and G.F.Y. analyzed the data; H.Y.S. and K.D.H wrote the paper; G.F.Y. and H.Z. interpreted the data and revised the manuscript.

Supplemental data

The following materials are available in the online version of this article.

Supplemental Figure S1. Determination of total anthocyanin contents and analysis of anthocyanin biosynthesis-related gene expression patterns in ‘Jinzheng NO.1’ and ‘Starkrimson’ pears at 30, 60 and 90 DAFB.

Supplemental Figure S2. The phenotype of T2 generation tomato plants with *ERF4.1* overexpression and deletion mutation.

Supplemental Figure S3. Identification of *ERF4.1* deletion in T2 generation tomato mutants.

Supplemental Figure S4. Interaction validation of PyERF4.1/PyERF4.2 with PyMYB114 or PybHLH3.

Supplemental Figure S5. Validation of the effect of PyMYB114 and PybHLH3 overexpression on the transcriptional activity of PyERF4.1/PyERF4.2.

Supplemental Figure S6. The amino acid sequences of PyERF4.2 and PyERF4.2 after EAR motif mutation (PyERF4.2M; Met residues mutated to Leu residues), with the EAR motif in red lettering.

Supplemental Table S1. RT-qPCR primers used in this study.

Supplemental Table S2. The primers used for vector construction in this study.

Funding

This work was supported by the National Natural Science Foundation of China (31901993, 31970312, 32272682, 32170315, 31970200), the Natural Science Foundations of Anhui Province (1908085MC72), the Key Research and Development Plan of Anhui Province (202003a06020011), and the Fundamental Research Funds for the Central Universities (JZ2021HGPA0063).

Conflict of interest statement. None declared.

References

An JP, Wang XF, Li YY, Song LQ, Zhao LL, You CX, Hao YJ. EIN3-LIKE1, MYB1, and ETHYLENE RESPONSE FACTOR3 act in a regulatory loop that synergistically modulates ethylene biosynthesis and anthocyanin accumulation. *Plant Physiol.* 2018;**178**(2):808–823. <https://doi.org/10.1104/pp.18.00068>

An JP, Wang XF, Zhang XW, Xu HF, Bi SQ, You CX, Hao YJ. An apple MYB transcription factor regulates cold tolerance and anthocyanin accumulation and undergoes MIEL1-mediated degradation. *Plant Biotechnol J.* 2020a;**18**(2):337–353. <https://doi.org/10.1111/pbi.13201>

An JP, Zhang XW, Bi SQ, You CX, Wang XF, Hao YJ. The ERF transcription factor *MdERF38* promotes drought stress-induced anthocyanin biosynthesis in apple. *Plant J.* 2020b;**101**(3):573–589. <https://doi.org/10.1111/tpj.14555>

Bai S, Tao R, Tang Y, Yin L, Ma Y, Ni J, Yan X, Yang Q, Wu Z, Zeng Y, et al. BBX16, a B-box protein, positively regulates light-induced anthocyanin accumulation by activating MYB10 in red pear. *Plant Biotechnol J.* 2019;**17**(10):1985–1997. <https://doi.org/10.1111/pbi.13114>

Castillejo C, Waurich V, Wagner H, Ramos R, Oiza N, Muñoz P, Triviño JC, Caruana J, Liu Z, Cobo N, et al. Allelic variation of MYB10 is the major force controlling natural variation in skin and flesh color in strawberry (*Fragaria* spp.) fruit. *Plant Cell.* 2020;**32**(12):3723–3749. <https://doi.org/10.1105/tpc.20.00474>

Chen H, Zou Y, Shang Y, Lin H, Wang Y, Cai R, Tang X, Zhou JM. Firefly luciferase complementation imaging assay for protein-protein interactions in plants. *Plant Physiol.* 2008;**146**(2):368–376. <https://doi.org/10.1104/pp.107.111740>

Choi HS, Seo M, Cho HT. Two TPL-binding motifs of ARF2 are involved in repression of auxin responses. *Front Plant Sci.* 2018;**9**(1):372. <https://doi.org/10.3389/fpls.2018.00372>

Cui D, Zhao S, Xu H, Allan AC, Zhang X, Fan L, Chen L, Su J, Shu Q, Li K. The interaction of MYB, bHLH and WD40 transcription factors in red pear (*Pyrus pyrifolia*) peel. *Plant Mol Biol.* 2021;**106**(4–5):407–417. <https://doi.org/10.1007/s11103-021-01160-w>

Deng H, Chen Y, Liu Z, Liu Z, Shu P, Wang R, Hao Y, Su D, Pirrello J, Liu Y, et al. *SIERF.F12* modulates the transition to ripening in tomato fruit by recruiting the co-repressor TOPLESS and histone deacetylases to repress key ripening genes. *Plant Cell.* 2022;**34**(4):1250–1272. <https://doi.org/10.1093/plcell/koac025>

Engler C, Marillonnet S. Combinatorial DNA assembly using Golden Gate cloning. *Methods Mol Biol.* 2013;**1073**(1):141–156. https://doi.org/10.1007/978-1-62703-625-2_12

Espley RV, Hellens RP, Putterill J, Stevenson DE, Kutty-Amma S, Allan AC. Red colouration in apple fruit is due to the activity of the MYB transcription factor, *MdMYB10*. *Plant J.* 2007;**49**(3):414–427. <https://doi.org/10.1111/j.1365-313X.2006.02964.x>

Fujimoto SY, Ohta M, Usui A, Shinshi H, Ohme-Takagi M. Arabidopsis ethylene-responsive element binding factors act as transcriptional activators or repressors of GCC box-mediated gene expression. *Plant Cell.* 2000;**12**(3):393–404. <https://doi.org/10.1105/tpc.12.3.393>

Han YC, Kuang JF, Chen JY, Liu XC, Xiao YY, Fu CC, Wang JN, Wu KQ, Lu WJ. Banana transcription factor *MaERF11* recruits histone deacetylase *MaHDA1* and represses the expression of *MaACO1* and *Expansins* during fruit ripening. *Plant Physiol.* 2016;**171**(2):1070–1084. <https://doi.org/10.1104/pp.16.00301>

Hichri I, Barrieu F, Bogs J, Kappel C, Delrot S, Lauvergeat V. Recent advances in the transcriptional regulation of the flavonoid biosynthetic pathway. *J Exp Bot.* 2011;**62**(8):2465–2483. <https://doi.org/10.1093/jxb/erq442>

Hichri I, Heppel SC, Pillet J, Léon C, Czemplak S, Delrot S, Lauvergeat V, Bogs J. The basic helix-loop-helix transcription factor MYC1 is involved in the regulation of the flavonoid biosynthesis pathway in grapevine. *Mol Plant.* 2010;**3**(3):509–523. <https://doi.org/10.1093/mp/spp118>

Hu Y, Han Z, Wang T, Li H, Li Q, Wang S, Tian J, Wang Y, Zhang X, Xu X, et al. Ethylene response factor *MdERF4* and histone deacetylase *MdHDA19* suppress apple fruit ripening through histone deacetylation of ripening related genes. *Plant Physiol.* 2022;**188**(4):2166–2181. <https://doi.org/10.1093/plphys/kiac016>

Jian W, Cao H, Yuan S, Liu Y, Lu J, Lu W, Li N, Wang J, Zou J, Tang N, et al. *SIMYB75*, an MYB-type transcription factor, promotes

- anthocyanin accumulation and enhances volatile aroma production in tomato fruits. *Hortic Res.* 2019;**6**(1):22. <https://doi.org/10.1038/s41438-018-0098-y>
- Kagale S, Links MG, Rozwadowski K. Genome-wide analysis of ethylene-responsive element binding factor-associated amphiphilic repression motif-containing transcriptional regulators in *Arabidopsis*. *Plant Physiol.* 2010;**152**(3):1109–1134. <https://doi.org/10.1104/pp.109.151704>
- Koes R, Verweij W, Quattrocchio F. Flavonoids: a colorful model for the regulation and evolution of biochemical pathways. *Trends Plant Sci.* 2005;**10**(5):236–242. <https://doi.org/10.1016/j.tplants.2005.03.002>
- Koyama T, Sato F. The function of ETHYLENE RESPONSE FACTOR genes in the light-induced anthocyanin production of *Arabidopsis thaliana* leaves. *Plant Biotechnol.* 2018;**35**(1):87–91. <https://doi.org/10.5511/plantbiotechnology.18.0122b>
- Lee HS, Wicker L. Anthocyanin pigments in the skin of lychee fruit. *J Food Sci.* 1991;**56**(2):466–468. <https://doi.org/10.1111/j.13652621.1991.tb05305.x>
- Lee JM, Joung JG, McQuinn R, Chung MY, Fei Z, Tieman D, Klee H, Giovannoni J. Combined transcriptome, genetic diversity and metabolite profiling in tomato fruit reveals that the ethylene response factor *SIERF6* plays an important role in ripening and carotenoid accumulation. *Plant J.* 2012;**70**(2):191–204. <https://doi.org/10.1111/j.1365-313X.2011.04863.x>
- Li C, Wu J, Hu KD, Wei SW, Sun HY, Hu LY, Han Z, Yao GF, Zhang H. *PyWRKY26* and *PybHLH3* cotargeted the *PyMYB114* promoter to regulate anthocyanin biosynthesis and transport in red-skinned pears. *Hortic Res.* 2020;**7**(1):37. <https://doi.org/10.1038/s41438-020-0254-z>
- Li LX, Wei ZZ, Zhou ZL, Zhao DL, Tang J, Yang F, Li YH, Chen XY, Han Z, Yao GF, et al. A single amino acid mutant in the EAR motif of *lbMYB44.2* reduced the inhibition of anthocyanin accumulation in the purple-fleshed sweetpotato. *Plant Physiol Biochem.* 2021;**167**(10):410–419. <https://doi.org/10.1016/j.plaphy.2021.08.012>
- Li T, Jiang Z, Zhang L, Tan D, Wei Y, Yuan H, Li T, Wang A. Apple (*Malus domestica*) *MdERF2* negatively affects ethylene biosynthesis during fruit ripening by suppressing *MdACS1* transcription. *Plant J.* 2016;**88**(5):735–748. <https://doi.org/10.1111/tpj.13289>
- Li X, Tao S, Wei S, Ming M, Huang X, Zhang S, Wu J. The mining and evolutionary investigation of AP2/ERF genes in pear (*Pyrus*). *BMC Plant Biol.* 2018;**18**(1):46. <https://doi.org/10.1186/s12870-018-1265-x>
- Liu H, Liu Z, Wu Y, Zheng L, Zhang G. Regulatory mechanisms of anthocyanin biosynthesis in apple and pear. *Int J Mol Sci.* 2021;**22**(16):8441. <https://doi.org/10.3390/ijms22168441>
- Liu Q, Kasuga M, Sakuma Y, Abe H, Miura S, Yamaguchi-Shinozaki K, Shinozaki K. Two transcription factors, DREB1 and DREB2, with an EREBP/AP2 DNA binding domain separate two cellular signal transduction pathways in drought- and low-temperature-responsive gene expression, respectively, in *Arabidopsis*. *Plant Cell.* 1998;**10**(8):1391–1406. <https://doi.org/10.1105/tpc.10.8.1391>
- Ma H, Yang T, Li Y, Zhang J, Wu T, Song T, Yao Y, Tian J. The long noncoding RNA *MdLNC499* bridges *MdWRKY1* and *MdERF109* function to regulate early-stage light-induced anthocyanin accumulation in apple fruit. *Plant Cell.* 2021;**33**(10):3309–3330. <https://doi.org/10.1093/plcell/koab188>
- Ni J, Bai S, Zhao Y, Qian M, Tao R, Yin L, Gao L, Teng Y. Ethylene response factors *Pp4ERF24* and *Pp12ERF96* regulate blue light-induced anthocyanin biosynthesis in 'Red Zaosu' pear fruits by interacting with *MYB114*. *Plant Mol Biol.* 2019;**99**(1–2):67–78. <https://doi.org/10.1007/s11103-018-0802-1>
- Ni J, Premathilake AT, Gao Y, Yu W, Tao R, Teng Y, Bai S. Ethylene-activated *PpERF105* induces the expression of the repressor-type R2R3-MYB gene *PpMYB140* to inhibit anthocyanin biosynthesis in red pear fruit. *Plant J.* 2021;**105**(1):167–181. <https://doi.org/10.1111/tpj.15049>
- Ohta M, Matsui K, Hiratsu K, Shinshi H, Ohme-Takagi M. Repression domains of class II ERF transcriptional repressors share an essential motif for active repression. *Plant Cell.* 2001;**13**(8):1959–1968. <https://doi.org/10.1105/tpc.010127>
- Peng M, Shahzad R, Gul A, Subthain H, Shen S, Lei L, Zheng Z, Zhou J, Lu D, Wang S, et al. Differentially evolved glucosyltransferases determine natural variation of rice flavone accumulation and UV-tolerance. *Nat Commun.* 2017;**8**(1):1975. <https://doi.org/10.1038/s41467-017-02168-x>
- Pirrello J, Jaimes-Miranda F, Sanchez-Ballesta MT, Tournier B, Khalil-Ahmad Q, Regad F, Latché A, Pech JC, Bouzayen M. *Sl-ERF2*, a tomato ethylene response factor involved in ethylene response and seed germination. *Plant Cell Physiol.* 2006;**47**(9):1195–1205. <https://doi.org/10.1093/pcp/pcj084>
- Qi Y, Yang Z, Sun X, He H, Guo L, Zhou J, Xu M, Luo M, Chen H, Tian Z. Heterologous overexpression of *StERF3* triggers cell death in *Nicotiana benthamiana*. *Plant Sci.* 2022;**315**(2):111149. <https://doi.org/10.1016/j.plantsci.2021.111149>
- Ramsay NA, Glover BJ. MYB-bHLH-WD40 protein complex and the evolution of cellular diversity. *Trends Plant Sci.* 2005;**10**(2):63–70. <https://doi.org/10.1016/j.tplants.2004.12.011>
- Tanaka Y, Sasaki N, Ohmiya A. Biosynthesis of plant pigments: anthocyanins, betalains and carotenoids. *Plant J.* 2008;**54**(4):733–749. <https://doi.org/10.1111/j.1365-313X.2008.03447.x>
- Wu T, Liu HT, Zhao GP, Song JX, Wang XL, Yang CQ, Zhai R, Wang ZG, Ma FW, Xu LF. Jasmonate and ethylene-regulated ethylene response factor 22 promotes lanolin-induced anthocyanin biosynthesis in 'Zaosu' pear (*Pyrus bretschneideri* Rehd.) fruit. *Biomolecules.* 2020;**10**(2):278. <https://doi.org/10.3390/biom10020278>
- Xu W, Grain D, Bobet S, Le Gourrierec J, Thévenin J, Kelemen Z, Lepiniec L, Dubos C. Complexity and robustness of the flavonoid transcriptional regulatory network revealed by comprehensive analyses of MYB-bHLH-WDR complexes and their targets in *Arabidopsis* seed. *New Phytol.* 2014;**202**(1):132–144. <https://doi.org/10.1111/nph.12620>
- Yan S, Chen N, Huang Z, Li D, Zhi J, Yu B, Liu X, Cao B, Qiu Z. Anthocyanin fruit encodes an R2R3-MYB transcription factor, *SIAN2*-like, activating the transcription of *SI-MYBATV* to fine-tune anthocyanin content in tomato fruit. *New Phytol.* 2020;**225**(5):2048–2063. <https://doi.org/10.1111/nph.16272>
- Yang J, Liu Y, Yan H, Tian T, You Q, Zhang L, Xu W, Su Z. PlantEAR: functional analysis platform for plant EAR motif-containing proteins. *Front Genet.* 2018;**9**(11):590. <https://doi.org/10.3389/fgene.2018.00590>
- Yang LP, Ling WH, Du ZC, Chen YM, Li D, Deng S, Liu Z, Yang L. Effects of anthocyanins on cardiometabolic health: a systematic review and meta-analysis of randomized controlled trials. *Adv Nutr.* 2017;**8**(5):684–693. <https://doi.org/10.3945/an.116.014852>
- Yang Y, Yao G, Yue W, Zhang S, Wu J. Transcriptome profiling reveals differential gene expression in proanthocyanidin biosynthesis associated with red/green skin color mutant of pear (*Pyrus communis* L.). *Front Plant Sci.* 2015;**6**(9):795. <https://doi.org/10.3389/fpls.2015.00795>
- Yao G, Ming M, Allan AC, Gu C, Li L, Wu X, Wang R, Chang Y, Qi K, Zhang S, et al. Map-based cloning of the pear gene *MYB114* identifies an interaction with other transcription factors to coordinately regulate fruit anthocyanin biosynthesis. *Plant J.* 2017;**92**(3):437–451. <https://doi.org/10.1111/tpj.13666>
- Zeng JK, Li X, Xu Q, Chen JY, Yin XR, Ferguson IB, Chen KS. *EjAP2-1*, an AP2/ERF gene, is a novel regulator of fruit lignification induced by chilling injury, via interaction with *EjMYB* transcription factors. *Plant Biotechnol J.* 2015;**13**(9):1325–1334. <https://doi.org/10.1111/pbi.12351>
- Zhai R, Wang Z, Zhang S, Meng G, Song L, Wang Z, Li P, Ma F, Xu L. Two MYB transcription factors regulate flavonoid biosynthesis in pear fruit (*Pyrus bretschneideri* Rehd.). *J Exp Bot.* 2016;**67**(5):1275–1284. <https://doi.org/10.1093/jxb/erv524>

- Zhang J, Xu H, Wang N, Jiang S, Fang H, Zhang Z, Yang G, Wang Y, Su M, Xu L, et al.** The ethylene response factor *MdERF1B* regulates anthocyanin and proanthocyanidin biosynthesis in apple. *Plant Mol Biol.* 2018;**98**(3):205–218. <https://doi.org/10.1007/s11103-018-0770-5>
- Zhao J, Dixon RA.** MATE transporters facilitate vacuolar uptake of epicatechin 3'-O-glucoside for proanthocyanidin biosynthesis in *Medicago truncatula* and *Arabidopsis*. *Plant Cell.* 2009;**21**(8): 2323–2340. <https://doi.org/10.1105/tpc.109.067819>
- Zhou H, Lin-Wang K, Wang F, Espley RV, Ren F, Zhao J, Ogutu C, He H, Jiang Q, Allan AC, et al.** Activator-type R2R3-MYB genes induce a repressor-type R2R3-MYB gene to balance anthocyanin and proanthocyanidin accumulation. *New Phytol.* 2019;**221**(4): 1919–1934. <https://doi.org/10.1111/nph.15486>